

Assembly Instruction

for High Frequency Spindle

MFW-1920/30/1 VC HSK-A63

Article No 114487



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Index O

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1. Introduction

These Assembly Instruction describe the mounting an connection procedures of spindle. These state-of-the-art spindles are constructed according to approved safety standards and regulations. Nevertheless during operation there is still an inherent danger to the user or third persons and at the same time damage to the machine or other property is possible. For this reason these Assembly Instruction must be read and understood by all persons who work with the machine.

1.1 Spindle Identification

In somecases the engraved number and type designation are no longer visible after the spindle has been installed. However, to ensure that the spindle can be identified later without any difficulty, we consult to print the 1st and the 3rd page from this Assembly Instruction and then enter the corresponding information.

Mounting Date: _____

Machine Number: _____

Starting up Date of Machine: _____

Machine Location: _____

1.2 Assembly Instruction: Target & Purpose

Target Readers

These Assembly Instruction are designed for:

- the **installation personnel** (mechanic, electrician, electronics technician) at the manufacturers of the machine in which the spindle is to be incorporated
- the **machine operator** and **service technician** at the end customer

The following **skills** have been assumed:

- Installation personnel
 - Mechanical work: training as mechanic (or equivalent training)
 - Electrical work: training as electrician or electronics technician (or equivalent training)
- Operator and service technician: training as mechanic (or equivalent training).

In addition it is assumed that these persons have the necessary basic knowledge on how to handle spindles

Purpose of the Assembly Instruction

These Assembly Instruction should enable the various target groups to carry out all working procedures safely and correctly. Therefore these Assembly Instruction must be read and understood by all persons who work on or with the spindle.

1.3 General Info on Using Assembly Instruction

Structure & Use

In these Assembly Instruction you will find all the necessary information of spindles, for their installation, operation, maintenance and for trouble shooting. This information is supplemented by relevant information on safety and hazards.

Information on the installation of periphery devices is not included in these Assembly Instruction. This information can be found in separate documents provided with the relevant devices.

All persons assigned to work on the spindle must read and understand the relevant chapters, and in particular the references to safety and hazards, before carrying out any work on or with the spindle.

In addition observe:

- the information and instructions in the documents for periphery devices
- all local safety regulations

Conventions Employed



This pictogram indicates important information for the correct and economical application/operation of the product.



This pictogram with an additional **"WARNING!"** at the beginning of the text indicates instructions on safety or hazards, which If ignored, **will result in danger to the user or third persons.**



This pictogram with an additional **"CAUTION!"** at the beginning of the text indicate instructions on safety and hazards, which If ignored, **could result in damage to the spindle or other property.**

- Procedures requiring **several steps** are numbered.
- **Lists** are indicated by a **dash "-"**.

Safekeeping of Documents

Always keep these Assembly Instruction (including documents for periphery devices) near to the associated machine so that they are always available.

Should you lose these Assembly Instruction, contact the company Fischer AG and for a nominal charge you will receive a replacement. Please send a fax containing the number of the spindle plus the language you require (German, English, French) to the address given further below.

Suggestions & Proposals for Modifications

We would welcome any comments and suggestions which could help to improve these Assembly Instruction. Should you have any suggestions and proposals for modifications, proceed as follows:

1. Copy the relevant page(s) and note your remarks or modifications.
2. Send the page(s) with your name and address (incl. telephone and fax number) to:

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2. Safety Precautions

2.1 Approved Applications

The spindles are designed and produced according to the latest trends and safety precautions. However, if not used according to instructions the spindles can cause hazards. The spindles are designed for high-speed milling works (HSC) and for the treatment on materials like aluminium, grey cast iron, steel, brass, titan, plastic, compound material, graphite and wood. The following indications for high-speed milling works have to be observed.



ATTENTION: In order to achieve a maximum life time of the spindle following important directions have to be observed. If you do not observe these directions, the spindle life time is reduced accordingly and the spindle could fail immediately.

Basic Directions

- Observe the balancing quality G 2.5 – Norm ISO 1940 – of the used tools.
- Observe the runout accuracy R2 of 0.01 mm at the end of the tool.
- Make sure a minimum oscillating speed of each toolholder with its tool and optimise the milling process.
- Use tools which are designed for High Speed Milling.
- Use tools with shafts as short as possible.
- Observe the tools are fixed rigidly in the toolholder, i.e. use shrinked or monolyte tools.

Displaying Picked-Up Power When Milling

- Register the picked-up power during the operation with each new tool (reference measurements).
- If the picked-up power during operation is higher than 30 % as the reference power, an alarm should be triggered.

Monitoring & Display of Spindle Vibrations in Idle and Milling Runs



Note! The vibration values in this instruction manual are indicated in the unit “peak calculated” (pc). Fig. 2-1 shows how the usual vibration values might be indicated.

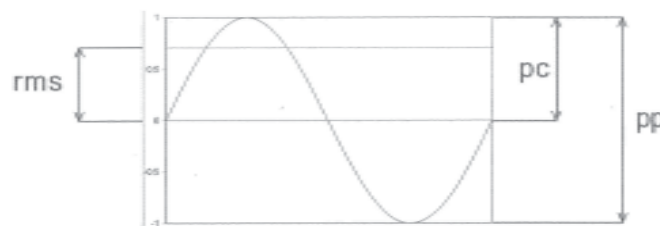


Fig. 2 - 1: Vibration values

- Checking the vibration while idling for max. rpm with the used and applied tools having a balance grade of G2.5. (Reference measurements) A max. value of <3mm/s pc is measured in the spectrum with the rotary frequency.
- Checking vibration during the milling process. Value <6mm/s pc, ensures max. service life of the spindle. Value of 8 - 10mm/s pc should not be exceeded during milling. Values >10mm/s pc represent peak values and may not be exceeded.

Any use exceeding these specifications is not permissible. Noncompliance with these rules results in damages. The company Fischer AG is not liable for any such damages.

The following are part of the intended use:

- Compliance with the rules, instructions, and notes of this instruction manual
- Compliance with the prescribed inspection and maintenance intervals
- Correct and proper spindle maintenance
- Use according to the prescribed environmental and operating conditions



CAUTION! Never use the spindle as centrifuge, fan drive, or drive of any type. This is clearly wrongful use and a violation of the intended use specifications. Such an act is excluded from the product liability of the company Fischer AG. The spindle as well as its components are destroyed by the excessive centrifugal forces when used for impermissible milling processes. Destroyed parts of the spindle are flung away from the spindle at a high velocity and can cause severe bodily injuries to persons in the proximity of the spindle.

These limitations can be waived after an appropriate investigation at Fischer AG.

2.2 Organizational Measures

- Only persons sufficiently qualified for the work involved and who have the necessary knowledge of the product are permitted to install, operate, service and if necessary repair the spindles.
It is the responsibility of the customer to ensure that the Assembly Instruction are supplemented by company-internal instructions with respect to supervision, reporting, work organization, staff qualification etc.
- The responsibility for the various tasks involved in the installation and operation of the spindle must be clearly determined and followed in order to prevent any doubt on accountability.
This applies in particular to work carried out on electrical, pneumatic and hydraulic equipment which demands specially trained personnel.
- All personnel engaged in working with the spindle **must have read and understood these Assembly Instruction.**
- Please observe the additions to the Assembly Instruction:
 - all warnings for safety and hazards contained in the separate documents for periphery devices.
 - all local and general instructions and regulations regarding work safety, prevention of accidents and protection of the environment.
- **Without written permission** obtained from Fischer AG, it is **not** permitted to carry out **modifications or extensions** to the spindles or periphery devices.
- For replacements to the spindles use **exclusively original accessories and spare parts** provided by Fischer AG.

2.3 Inherent Product Hazards



WARNING! Even when used strictly according to instructions, when working on or with the spindle an inherent danger to life and limb of the user or third persons still exists and damage can be caused to the spindle or other property. Therefore the following **safety and hazard warnings must be strictly observed**:

- The spindles have a very high rotational speed. Therefore in operation, splinters or, should a tool break, pieces of tool can fly out at very high speed. The following measures must be carried out on the machine in which the spindle is installed:
 - It must be possible to enclose the working space in a cabin.
 - The door of the cabin must be monitored by a technical safety arrangement so that the spindle can only be operated when the door is closed.
 - The cabin must be constructed so that splinters and pieces of tool cannot penetrate the walls or windows of the cabin.

WARNING! If this warning is ignored, persons could be injured or killed by flying splinters or pieces of tool.

- Under no circumstances are the fingers to be placed inside the taper mount.

WARNING! If this warning is ignored, the fingers could be crushed.

- Due the built-in spring nest, individual components of the spindle are under high tension. If the machine is incorrectly disassembled, these components can fly out at high speed. Therefore only properly trained personnel are permitted to open the spindles.

WARNING! If this is ignored, persons can be injured or killed by flying parts of the spindle.

- All tools and tool holders must meet the standard EN ISO 15641 (centrifugal force test) and be dynamically balanced according to the quality level G2.5 set out in the standards VDI 2060 or ISO 1940. The maximum permitted rotation speed of the tool/tool holder and the spindle is not to be exceeded under any circumstances.

WARNING! If this is ignored, persons can be injured or killed by flying parts of the tool or tool holder.

- Due the switching tolerance of the proximity switches, there can be no 100% guarantee that the tension of the tool is absolutely correct. This is only possible using a tool facing stop system.

- In operation, the spindles are monitored by proximity switches in order to provide protection against inadmissible operating conditions. It is not permitted to remove or bypass these proximity switches.

WARNING! If not observed there is a possibility that:

- the tool is not correctly clamped and it could therefore fly out (danger to persons).
 - spindles with HSK interface without a tool holder could start to rotate and thus damage the clamping system.
 - the spindle will be damaged due to operating conditions which are not permitted.
- The springs of the tool clamping system are subject to metal fatigue. For this reason their clamping power must be checked at least every six months. If the clamping power **falls below the minimum permitted power** (see paragraph 6.3), **the springs must be replaced.**

WARNING! If not observed, the tool clamp can be torn out during operation (danger to persons and property).

2.4 Additional Hazards

Electric System

- Only skilled and fully trained personnel who are familiar with the inherent dangers are permitted to work on the electrical system. In cases where the machine has to be worked on under power, an additional skilled person should be called in who can take the necessary steps in emergency situations.
- Before carrying out any work on the electrical system:
 - disconnect the machine and periphery devices from the mains and ensure that the power cannot be reconnected.
 - wait until the discharge time of the frequency converter has elapsed (see the instruction for the frequency converter).

Hydraulic System (if applicable)

- Work on the hydraulic system is only to be carried out by skilled personnel.
- Before carrying out any work on the hydraulic system, switch off the machine and the hydraulic power unit, if separately powered, and ensure that the power cannot be inadvertently switched back on again. Afterwards ensure that the hydraulic system is no longer under pressure (see the documentation for the hydraulic power unit).

Pneumatic System

- Work on the pneumatic system is only to be carried out by skilled personnel.
- Before carrying out any work on the pneumatic system, switch off the machine and the pneumatic set, if separately powered (close valve) and ensure that it cannot be switched back on inadvertently. Afterwards ensure that the pneumatic system is no longer under pressure (see the documentation for the pneumatic periphery devices).

Cooling System

- Work on the cooling system is only to be carried out by skilled personnel.
- Before carrying out any work on the cooling system, switch off the machine and the cooling system, if powered separately, and ensure that they cannot be switched back on inadvertently (see documentation for the cooling system).

Fuels

- Fuels can be hazardous to persons in several ways (allergic reaction, danger of slipping if the material leaks on the floor, etc.). Therefore take great care when handling such materials and always observe the relevant factory safety regulations.

2.5 Protect Environment

When not correctly used, Fuels can pollute the environment. Therefore handle such materials with care and strictly observe local regulations and laws.

Collect the used materials in suitable containers (e.g. original canisters) and ensure that the correct disposal procedures are carried out.

3. Spindle Description

3.1 Functioning

The function of the spindles is as follows:

The spindle shaft is driven by a synchronous/asynchronous motor, built into the spindle sleeve. The rotation speed of the motor is continuously variable via the static frequency converter. The specified frequency parameters can be found in chapter 10 "Annex".

The spindle shaft is fixed in the spindle sleeve by two to five bearings. Depending on the model of the spindle, the bearings are lubricated by grease or the oil-air lubrication system.

The oil/air mixture of the oil-air lubricated spindles is injected radially, i. e. directly into the most highly-stressed bearing zone. (DLS = Direct Lubrication System). The spindles without oil/air lubrication system are greased for life ex works.

The DLS system drains the interior of the spindle from excessive oil. Depending on the model of the spindle the excessive oil is sucked off by a vacuum pump.

The cooling system is integrated in the spindle sleeve. The spindle is liquid cooled to keep the motor and the bearings at the right operation temperature. It also keeps the longitudinal expansion of the spindle shaft as low as possible. Circulating coolant absorbs the heat lost by motor and it is then cooled itself on the way through an external cooler.

The protecting air system builds up an overpressure in the spindle. A special labyrinth seal prevents the penetration of dirt and humidity.

The tool clamping system is driven hydraulically and is integrated in the spindle shaft. It is provided with a HSK- or an ISO-Interface. If not by hand (HSK, collet, threading) the tools are clamped by the force of a spring unit and unclamped by a hydraulic cylinder.

An alternative unclamping unit (if any) consists of a hydraulic cylinder for the tool clamping system and a retention device for the spindle shaft during the unclamping procedure.

The thru-coolant unit (if any) is routing the cooling lubricant through the spindle shaft to the tool. It is integrated in the unclamping unit (if any) and functions as the interface between the rotating spindle shaft and the stable spindle sleeve.

The proximity switches monitor the automated tool clamping process to the machine controls. They check the positions of the tool clamping system ie clamped, unclamped or clamped without tool.

Another solution can be one or two analog sensors to monitor and control the tool clamping system. The sensors generate a current corresponding to the unclamped, clamped-with-tool or clamped-without-tool positions. The current is then processed by the sensor controller that sends its signals to the machine controls.

Zero-Speed Sensor S1 monitors the standstill of the spindle shaft and / or the encoder measures and monitors the position and the rotation speed of the spindle shaft.

The temperature gauges (if any) control the bearing and the stator temperature. The values are transmitted and processed by the machine control system.

The taper blowout system (if any) cleans the taper during the procedure of the tool change.

3.2 Range of Application

The spindle can be used for:

- **High-speed cutting (HSC)** of materials like aluminium, grey cast iron, steel, brass, titan, plastic, compound material, graphite and wood.



Any other application must be consulted with Fischer AG.

3.3 Standard Delivery

The standard delivery includes:

- Spindle
- Assembly Instruction manual
- Wooden box

3.4 Hazardous Areas & Safety Features

Hazardous Areas

- Rotating parts of spindle and tool
- The interface between spindle and tool clamp
- Electric connections
- Hydraulic connections
- Pneumatic connections
- Motor coolant connections

Safety Features

In order to guarantee safe operation, spindles and their periphery devices are equipped with several monitoring devices:

- Sensors integrated in the spindles
 - 1-2 sensor for monitoring the temperature of the spindle motor
 - 0-3 sensor for monitoring the temperature of the spindle bearing (PT100)
 - 1-5 proximity switches for tool clampingor
 - 1-2 analogue sensor for tool clamping
 - 1 zero-speed sensor and / or encoder
- The following monitoring systems must be integrated into the peripheries (if any):
 - in the motor coolant system
 - Temperature sensor
 - Flow detector
 - in the oil/air lubrication system
 - Pressure sensor
 - Oillevel control
 - Pump control
 - in the frequency converter
 - Current limiter
 - PTC, KTY or K227 triggering

3.5 Tooling Requirements & Safety

Toolholders

The following point must be observed in connection with the tool holders used with spindles:

- The tool holder should be as **short and light** as possible and **symmetrical** whenever possible.
- All tool holders must be dynamically balanced according to **quality rating G2.5** listed in the **standards VDI 2060 or ISO 1940**.
- **HSK tool holders** have facing systems and must be adjusted according to DIN 69 893. The axial preload dimension must be 0.02 - 0.08 mm.



CAUTION! On **ISO tool holders** the retention knobs must be screwed in flush and secured with "Loctite medium" (or an equivalent product).

Tools

- **Maximum eccentricity: 0.01 mm**
- Tools must be dynamically balanced to **quality rating G2.5** listed in standards **VDI 2060 or ISO 1940**.
- The maximum rotation speed must be engraved on all tools.
- In order to achieve a high metal removing rate, shank-type cutters must be clamped in a hydro-expansion chuck or monolith cutting tools will have to be used.



WARNING!

Tool holders and tools which do not meet the above-listed specifications are not be used with the spindles.

Tool holders and tools must each be subjected to a centrifugal force test according to EN ISO 15641.

The maximum permitted rotation speed of the spindle is not to be exceeded.



Information! When you are uncertain regarding critical rotations speeds, you can rely on tools and spindles obtained from Fischer AG.



Note!

Tools for a spindle with thru-coolant must be fitted with a transfer tube. For details, see Paras 3.5.1, 3.5.2 and 3.5.3 below.

3.5.1 HSK Toolholders for Spindles with Thru Coolant



Caution!

Spindles with HSK interface and thru coolant must be operated only with the toolholders fitted with a thru-coolant transfer tube.

Operating a toolholder without a transfer tube may cause damage to the spindle.

3.5.2 Fitting Transfer Tubes in Toolholders for Spindles with Thru Coolant

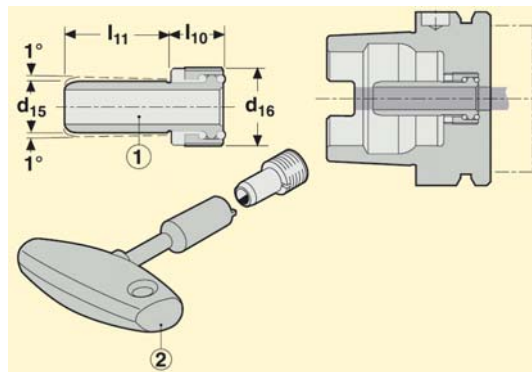


Fig 3-1: Thru-Coolant Transfer Tube

Legend:

- 1. Transfer tube
- 2. Transfer tube fitting key



Info: For tube tightening torques and use/not use of Loctite, see the instructions of your toolholder supplier.

Use only a transfer tube compatible with your tool interface. Once fitted, a tube must be flexible by $\pm 1^\circ$ at its fixing point to prevent its damage and jamming in the clamping set and also to prevent damage to the guide or sealing lips.

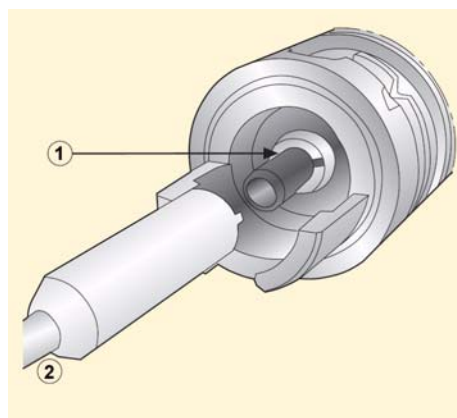


Fig 3 - 2: Correct Fitting of Thru-Coolant Transfer Tube

After fitting in, the tube must allow $\pm 1^\circ$ flexing.

3.5.3 HSK Toolholders for Spindles without Thru Coolant



Caution! For spindles with HSK interface (and axial bore) and no thru coolant make sure that no coolant penetrates the clamping set through an open tool or collet.

An open tool or collet operated with a HSK taper can cause a break-down of the tool interface at an early stage by inviting external coolant and dirt particles into the HSK clamping set.

When using a plug, make absolutely sure that it will not get loose.

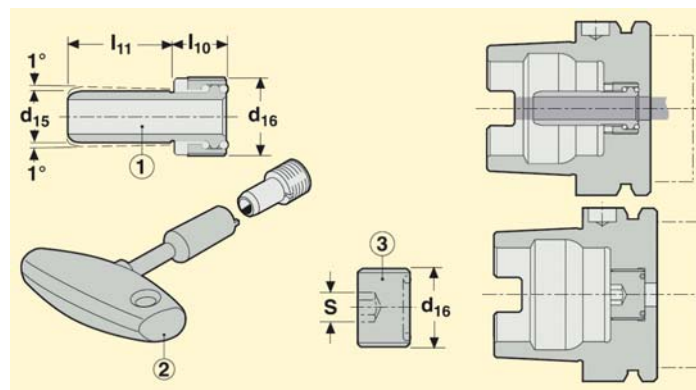


Fig 3 - 3: Transfer Tube or Plug



Info: Hence, for a spindle with no thru coolant and operated with an open tool or collet, secure the HSK taper either with an original transfer tube or an original plug.

For plug tightening torques and use/not use of Loctite, see the instructions of your toolholder supplier.

For fitting a transfer tube, see Para.3.5.2 above.

4. Installation & Commissioning

4.1 Installation & Commissioning Safety



WARNING! The following guidelines on safety and hazards are to be strictly observed and implemented. If ignored, it could result in danger to persons and/or damage to the spindle or other property.

- Only persons sufficiently qualified for the particular work and who have the necessary knowledge of the product are permitted to install and commission the spindles.
- The customer is responsible for assigning personnel for the various tasks associated with the installation and commissioning of the spindles and for ensuring that they are **qualified** and have the **necessary competence**.
- All persons assigned the task of carrying out installation and commissioning work must have read and understood these operating instructions.
- Without the written permission of Fischer AG, it is **not permitted to carry out any extensions and modifications** to the spindle or to any periphery devices supplied by Fischer AG.
- All safety equipment (proximity switches) are to be connected to machine control according to the relevant function. The safety equipment is **not to be bridged or removed**.

4.2 Checks Before Installation

Before carrying out the installation procedures make the following checks:

- **Equipment supplied**

- Check that the equipment supplied conforms to that listed on the delivery note.
- Check for any transport damage (any damage should be reported within 10 days of receipt to the transport company and to the supplier).

- **Spindle mounting in the machine**

The spindle mount must meet minimum specifications, which can be found in chapter 9 "Technical specifications".

- **Peripherals**

Various periphery devices are required for the operation of the spindle. Check whether all the periphery devices listed below are available and whether they meet minimum specifications, which can be found in chapter 9 "Technical specifications".



Note! The installation of the periphery devices is not dealt with in these operating instructions. For the necessary information consult the relevant documentation supplied with each periphery device.

4.3 Mounting the Spindle

Safety Precautions



- **WARNING!** Use only hoisting equipment with sufficient bearing capacity and always observe the basic safety regulations regarding transport.

Information: The weight of your spindle can be found in paragraph 9.1 "Technical specifications".

- **WARNING!** The spindle may only be secured on its sleeve using a hemp rope or a clamping system. It is assumed that persons assigned the task of transporting the spindle have adequate knowledge on how to secure loads correctly.
- **CAUTION!** When installing (and removing) or shifting the spindle in its holder, it is only permitted to strike the spindle sleeve. When doing this only use a plastic hammer.

Striking the spindle shaft can cause damage to the spindle.

Mounting position

Your spindle can be mounted either **horizontally or vertically** (mount facing downwards).

Preparation for mounting

1. Lift the spindle out of its wooden box by hand or with the aid of a crane (weight >25 kg) and place it on a soft surface (wood, cloth, cardboard etc.).
2. Remove the anti-corrosion paper and clean the spindle sleeve.
3. Smear a little oil or grease on the spindle sleeve in order to protect the contact surface in the clamping area (spindle sleeve/spindle holder) from corrosion.
4. Clean the spindle holder attachment hole in the machine.

Installation

Please refer to the operating instructions supplied with your machine tool when installing the spindle. The operating instructions will give you exact details and information on the precautions required.



ATTENTION! When installing the spindle, ensure that

- the spindle fixing dimensions are adhered to
- the axes and the alignment of the spindle respectively and the tool mount of the tool changer align precisely in the changing position.
- the stroke of the tool changing sequence matches exactly the spindle position.

4.4 Installation of Peripherals



Note: In Chapter 9 "Technical specifications" and in Chapter 10 "Annex" you will find all the necessary information and diagrams for the installation and the commissioning of the spindle. Further information and technical specifications can be found in the documentation of the periphery units.

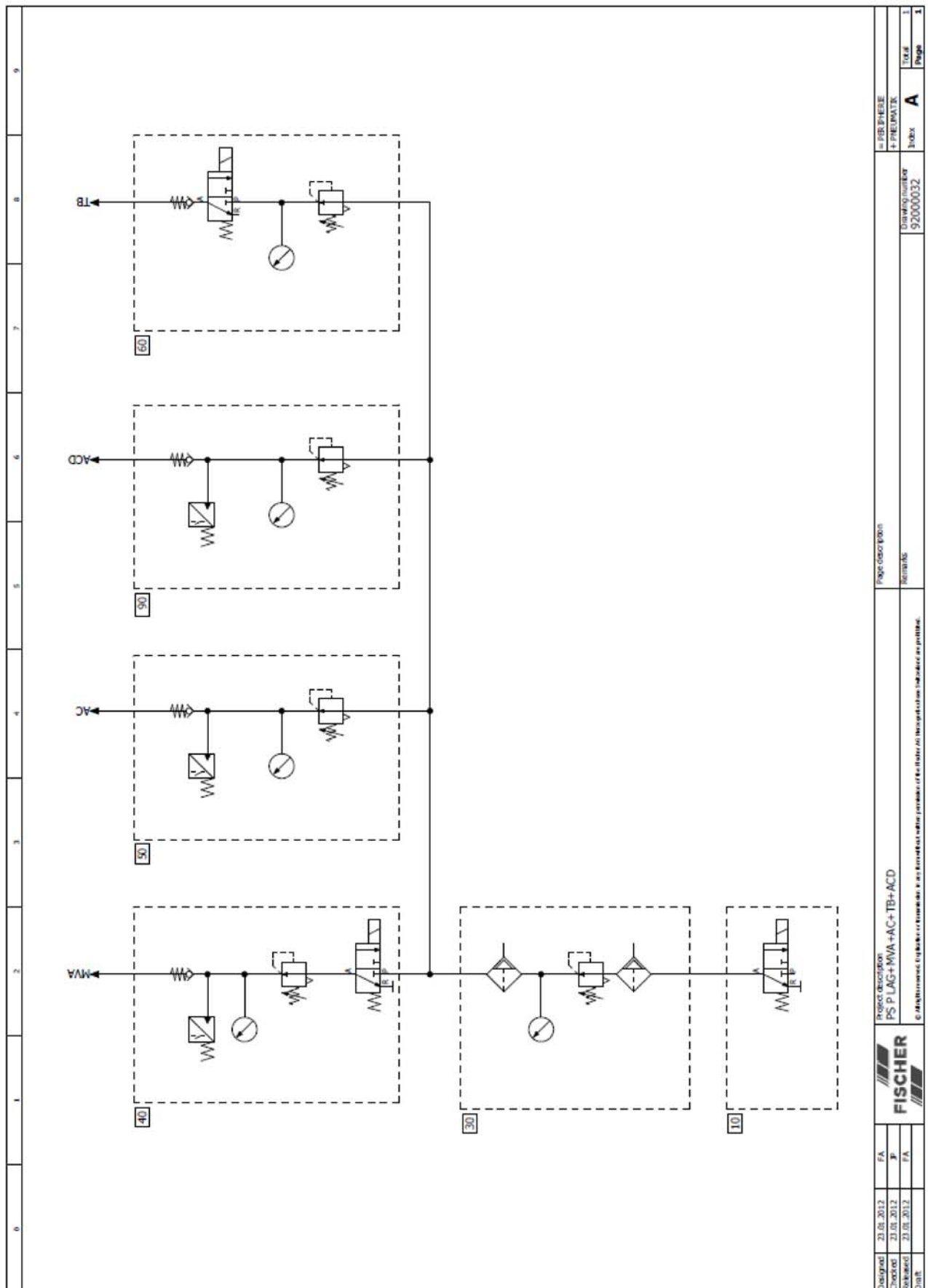
4.4.1 Installation of Pneumatic System

Pneumatic Installation: Schem 92000032, see next page

Compressed air, according to ISO standard 8573.1 (class 4.4.4) must be feed to the air treatment unit, before it is supplied to the different systems of the spindle.

Legend:

10	Main check valve
30	Air treatment unit
40	Lubricating air
50	Air curtain
60	Taper blowout
90	Air curtain for coolant trough unit
MVA	Oil-air lubrication
AC	Air curtain
TB	Taper blowout



Connection Specifications

For the connection of the air treatment unit, follow Technical Specifications in Chapter 9.



CAUTION! Unless an air treatment unit is used which meeting the minimum specifications given in Chapter 9, the spindle can be damaged.



Information: Further information about the connecting data of the air treatment unit can be found in a separate manual of the air treatment unit.

Connection Procedures

Connect the lines of **the oil-air lubrication, curtain air flow and taper blowout** to the spindle and to the air treatment unit acc. to Pneumatics Installation Schematic. Observe the following remarks:

- Observe the installation and service instructions in the documentation of your air treatment unit.
- Respect the absolute cleanness during the installation procedure.
- Keep all the connectors of the spindle closed, by means of blind caps, until you finally connect all the lines.
- Install all the periphery units in consideration of the length of lines and prepare the lines accordingly.
- The external diameter of the lines must correspond to the line specifications given in Chapter 9. The internal diameter should be as wide as possible in order to guarantee a maximum flowrate.
- The lines must not be under tension.
- Place the tube connections in a cable duct or secure them to suitable points at the machine.
- The lines must not be bent, respect minimal bend radius during installation of the lines.
- The lines must stand the required pressure and must have the necessary quality.



Caution! The dynamic pressures P2 must be observed. After replacing components and changes of pressure within the air supply system, the dynamic pressure P2 must be checked and readjusted 15cm before the spindle connector. If the pressure is not correctly set the spindle could be damaged and breaks down.



Note! Observe the minimum diameters of the valves in consideration of the minimum internal diameters of lines.

4.4.2 Connecting Hydraulic Tool Clamping System

Installation Schematic

The following drawing shows the installation for the hydraulic operated tool clamping system.

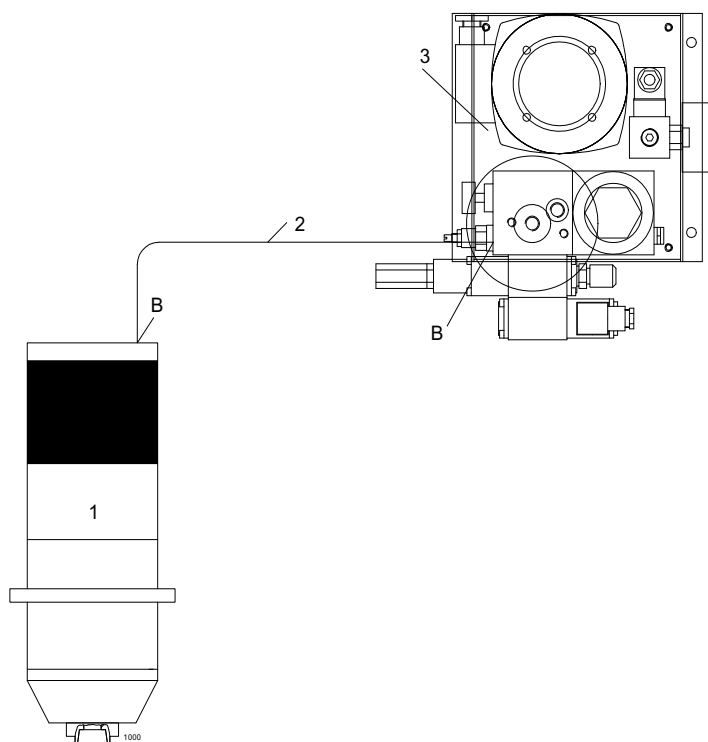


Fig. 4 - 3: Schematic of Hydraulic Tool Clamping System

Legend

- 1 Spindle
- 2 Hydraulic tubes
- 3 Hydraulic power unit
- B Tool unclamping

Connection Specifications

For the connection of the hydraulic system follow Technical Specifications in Chapter 9.



Information: For more information on the connection data to the hydraulic power unit consult the separate operating instruction supplied with the hydraulic power unit.

Connection Procedure

Connect the hydraulic power unit to the spindle according to Fig. 4 - 3 and observe the following remarks:

- Observe the installation and service instructions in the documentation of your hydraulic power unit.
- Respect the absolute cleanliness during the installation procedure.
- Keep all the connectors of the spindle closed, by means of blind caps, until you finally connect all the lines.
- Install all the periphery units in consideration of the length of lines and prepare the lines accordingly.
- The external diameter of the lines must correspond to the line specifications given in Chapter 9. The internal diameter should be as wide as possible in order to guarantee a maximum flowrate.
- The lines must not be under tension.
- Place the tube connections in a cable duct or secure them to suitable points at the machine.
- The lines must not be bent, respect minimal bend radius during installation of the lines.
- The lines must stand the required pressure and must have the necessary quality.



Information: The shorter the tubes the quicker the build-up of pressure for tool changing. When the distance between the spindle connection and the hydraulic power unit is greater than 10m, it is advisable to separate the valve block together with pressure accumulator and valve from the hydraulic power unit and install them as near as possible to the spindle.

4.4.3 Connecting Spindle Motor Cooling System without Secondary Cooling

Installation Schematic

The following drawing shows the installation of the cooling system.

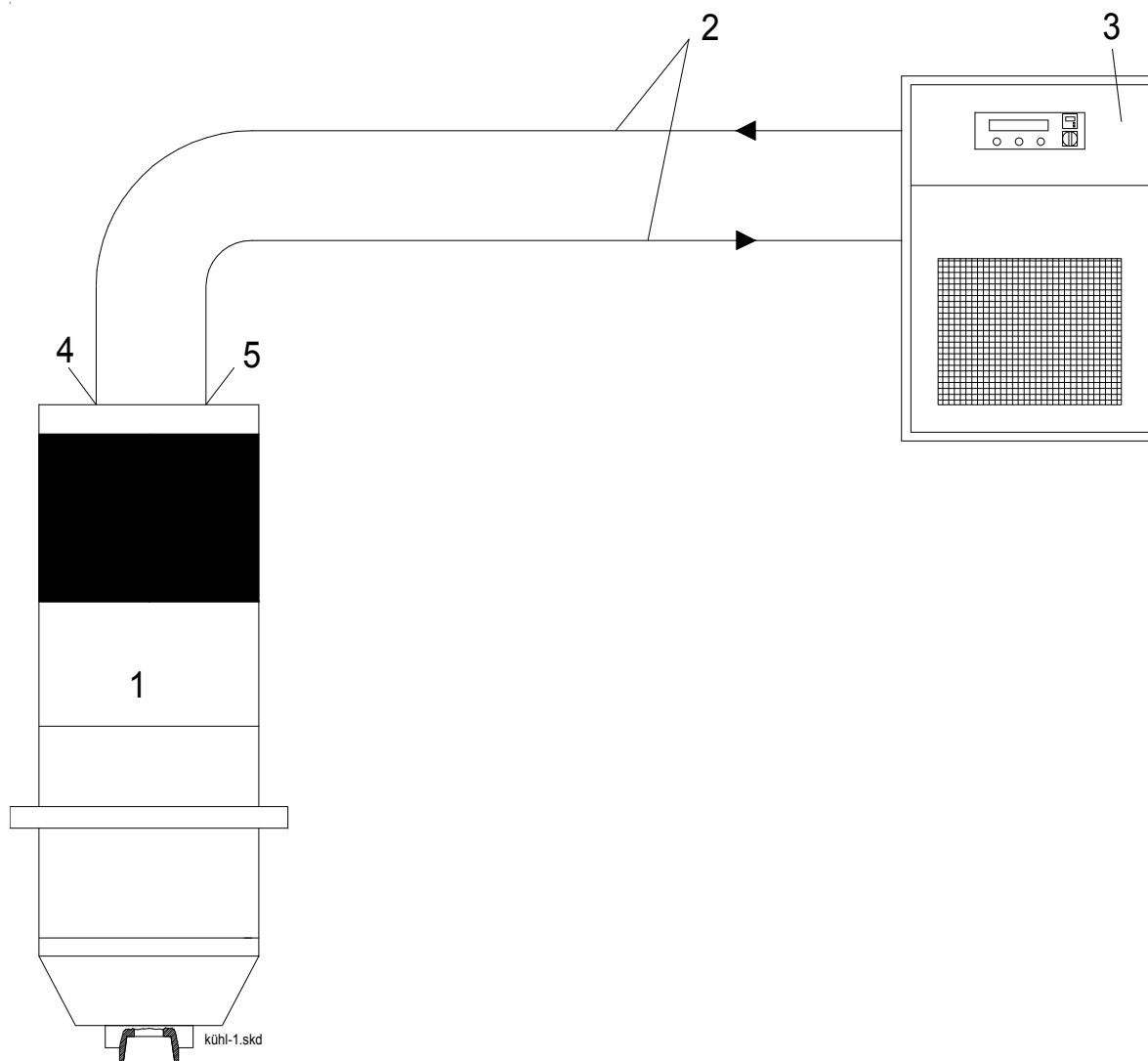


Fig. 4 - 4a: Cooling System Schematic

Legend

- | | |
|----------------|-----------------------|
| 1 Spindle | 4 Coolant IN fitting |
| 2 PEUR tubes | 5 Coolant OUT fitting |
| 3 Cooling unit | |

Connection Specifications

For the connection of the cooling system, follow Technical Specifications in Chapter 9.



Note: For more information about the motor cooling system consult the separate operating instructions supplied with the cooling unit.

Connection Procedure

Connect the spindle to the motor cooling system according to the drawing in Fig. 4 - 4a and observe the following remarks:

- Observe the installation and service instructions in the documentation of your cooling unit.
- Respect the absolute cleanliness during the installation procedure.
- Keep all the connectors of the spindle closed, by means of blind caps, until you finally connect all the lines.
- Install all the periphery units in consideration of the length of lines and prepare the lines accordingly.
- The external diameter of the lines must correspond to the line specifications given in Chapter 9. The internal diameter should be as wide as possible in order to guarantee a maximum flowrate.
- The lines must not be under tension.
- Place the tube connections in a cable duct or secure them to suitable points at the machine.
- The lines must not be bent, respect minimal bend radius during installation of the lines.
- The lines must stand the required pressure and must have the necessary quality.
- * Unless the exclusively recommended and ready-for-use COOL-X coolant is used a water-miscible corrosion inhibitor must be added to the cooling water to protect the spindle from corrosion and to keep the coolant biostable.
- * Unless the exclusively recommended and ready-for-use COOL-X coolant is used add from the beginning the appropriate additive according to Chapter 9: Technical Specifications in order to prevent organic residues (fungus).
- * Filter the coolant to keep it clean. For this you need a fine filter according to Chapter 9: Technical Specifications.



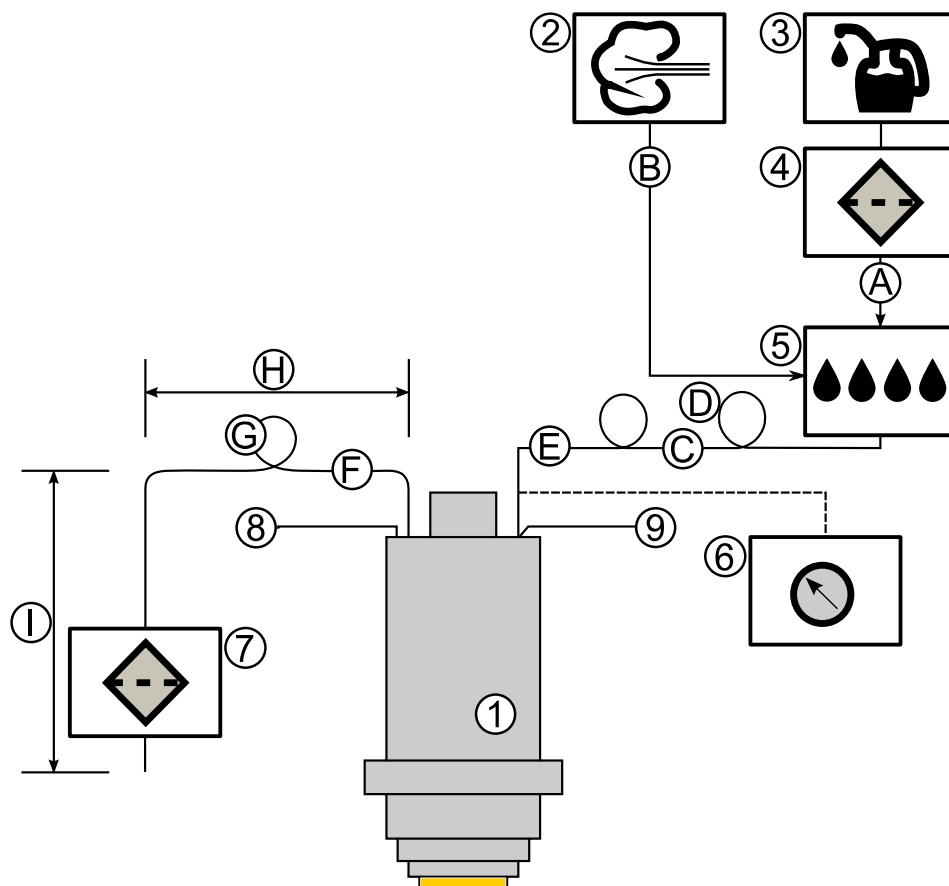
Note: Unless the installed lines are made of 100% PEUR, the life time of the lines can be reduced because of holes.

* This does not concern spindles cooled with oil.

** Unless filtered, the coolant can cause accumulation of mud and impurities in the coolant conduits and their clogging up.

4.4.3 Installation of the lubrication system

Installation overview



1	Spindle	A	Lubricating oil flow
2	Compressed air supply	B	Air supply line
3	Lubricating oil pump	C	Distance between coils
4	Fine flow filter	D	Oil reservoir; coil leader
5	Dosing unit	E	Distance from the first coil to the spindle
6	Pressure gauge	F	Oil-air lubrication return flow
7	Oil separator	G	Backstop; oil required by the coil
8	MV1; oil-air lubrication connection 1	H	Length of the spindle lubrication return
9	MVX; oil-air lubrication connection X	I	Height of the spindle lubrication return

Installation of the lubrication system

Preparatory measures



Note

The lubricating oil lines may only be connected to the spindle after the oil-air lubrication system has completed its flushing process!



Note

Be sure to read the product documentation for the lubricating oil pump before installation!



Note

Be sure to read the product documentation for the oil pressure monitor before installation.



Note

Be sure to read the product documentation for the air pressure monitor before installation.

Flushing of the lubrication system

This ensures that any possible impurities and preservatives in the dosing unit and in the lines are flushed out.

Procedure

1. Fill the lubricating oil pump's media container to half way
2. Remove the stoppers from the free ends of the oil-air lines
3. Place the open ends of the lines in a collection container
4. Turn the oil-air lubrication system on
5. Open the bleed screw until the oil is discharged without bubbles
6. Close the bleed screw
7. Leave the oil-air lubrication system running for 4-5 hours

Installation of the oil-air lubrication lines

Procedure

1. Remove the plugs from the spindle's oil lubrication connections
2. Connect the oil lubrication lines

Installation of the oil return line



Note

The oil return lines should always be aligned downwards!

Procedure

1. Remove the oil return line end cap at the spindle
2. Connect the line between the oil return line outlet and the oil separator
3. Maintain a maximum line length of 3m between the spindle and the oil separator
4. Maintain a maximum vertical range of 1m

Setting the oil and air pressure



Note

Setting the wrong oil-air pressure may result in damage to or even failure of the spindle!



Note

In order to measure the dynamic pressure while setting the oil lubricant pressure, the spindle shaft must be turned by hand!



Note

Once the dynamic pressure has been set at the lubrication connections, the corresponding manometer must be removed from the line!

Procedure

1. Connect a manometer to the lubrication line 15cm away from the spindle
2. Set the dynamic pressure for each individual lubrication connection to 4bar
3. Set the pressure of the compressed air supply to the dosing unit to between 4 - 5bar

Setting the quantity of lubricant

The spindle requires different quantities of lubricant for the different operation conditions:

- Continuous operation (**operational lubrication**)
- First operational start-up and after a long period of spindle down time (**pre-lubrication**).

Adjusting the quantity of lubricant

Adjustment of the quantity of lubricant for the appropriate operating conditions occurs through:

- Amendment of the lubricating oil pump's pause time (lubrication cycle times).
- A lubrication cycle consists of pump running time and pause time.



Note

The adjustment of the quantity of lubricant for the required operating conditions set out in this manual is based on changing the pause time!



Note

The adjustment of the quantity of lubricant set out in this manual is designed for a dosing unit with a dosage volume of 10mm³!



Note

Oscillation of the manometer needle during measurement of the dynamic pressure is an indication that the lines have been laid correctly.

Setting the lubrication cycle times

The lubrication cycle time for **pre-lubrication** is 30 seconds.

The lubrication cycle time for **lubrication** is 2 minutes.

Setting the pump advance running time

The pump advance running time is dependent on the length of the lines between the lubricating oil pump and the dosing unit.

The pump advance running time is:

- 10 seconds for line lengths of up to 10 metres.
- 15 seconds for line lengths between 10 metres and 15 metres.
- 20 seconds for line lengths between 15 metres and 40 metres.



Note

In order to monitor the drop in pressure, an external pressure gauge must be mounted to the dosing unit for line lengths between 15 and 40 metres!

Installation optimisation**Optimisation in the lubricating oil supply line**

In order to be able to reduce the pre-lubrication times required after switching the lubrication system on again, coiled lines are recommended. Oil reserves build up in these which quickly supply the bearings with lubricant again and prevent the oil from flowing back when the lubrication system is turned off.

Setting the pre-lubrication time**Note**

The pre-lubrication time is 3 minutes per metre of line length between two oil reservoirs!

Procedure

1. Lay out the first coiled line approx. 30cm before the spindle
2. Always align the coiled hose vertically or inclined at a maximum angle of 30°

Optimisation in the lubricating oil return line

In order to prevent the used lubricating oil from flowing back into the spindle when the lubrication system is turned off, you can use coiled lines in the return line.

Procedure

1. Create a return stop in the oil return line near the spindle by forming a coil
2. Always align the coiled hose vertically or inclined at a maximum angle of 30°

Controlling the oil-air lubrication system

1. The oil-air lubrication system should be turned off during any spindle shutdown periods of more than 20 seconds.
2. The shutdown period for the spindle must be queried through the controls on each start-up of the spindle.
3. Three different release scenarios for the start-up of the spindle must be set up in the controls.

Release scenario 1

Following down time of less than one hour.

Process:

1. Start lubrication
2. Release spindle start

Release scenario 2

For spindle down times of more than 1 hour.

Process:

1. Start lubrication
2. Activate warm-up process 1
3. Release spindle start

Release scenario 3

For spindle down times of more than 4 hours.

Process:

1. Activate pre-lubrication
2. Start lubrication
3. Activate warm-up process 2
4. Release spindle start

Warm-up process 1

After shutdown periods of more than one hour, the spindle must be warmed up in three periods of running time, at 50%, 75% and 90% of the maximum rotational speed, of one minute each.

Warm-up time optimisation

The warm-up procedure can be shortened for operational speeds which are lower than the maximum rotational speed

Operating speed	Warm up → Spindle start authorisation
less than 50% of the max.	No warm up required
less than 75% of the max.	after one minute for 50% of the maximum rotational speed
less than 90% of the max.	after one minute each for 50%, 75% of the maximum

Warm-up process 2

After shutdown periods of more than four hours, the spindle must be warmed up in three periods of running time, at 50%, 75% and 90% of the maximum rotational speed, of two minutes each.

Warm-up time optimisation

The warm-up procedure can be shortened for operational speeds which are lower than the maximum rotational speed.

Operating speed	Warm up → Spindle start authorisation
less than 50% of the max.	No warm up required
less than 75% of the max.	after two minutes for 50% of the maximum rotational speed
less than 90% of the max.	after two minutes each for 50%, 75% of the maximum



Note

The warm-up procedure must always be carried out following shutdown periods of more than four hours.

Monitoring of the oil-air lubrication system

1. The oil level in the lubrication oil pump must be monitored using a level regulator. Alternatively, or additionally, a flow monitor can be installed in the dosing unit's oil supply line.
2. The pressure of the supply of compressed air to the dosing unit must be monitored using a pressure gauge.
3. In order to ensure that the spindle is not damaged, it must be turned off by means of appropriate programming in the controls in the event that one of these monitoring components registers an error.

4.4.5 Connecting Thru-Coolant / MQL System for Spindles with Rotary Union

Installation Schematics

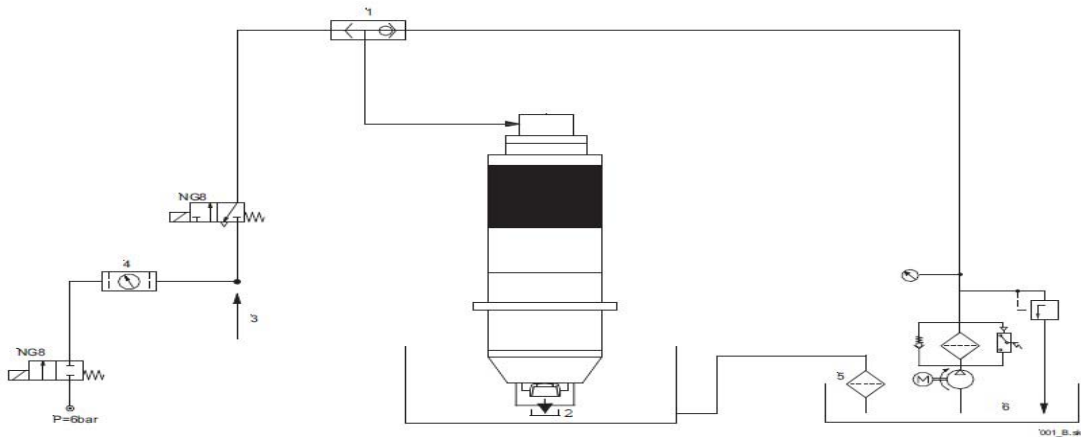


Fig. 4 - 6a: Schematic of Thru-Coolant System

Legend

- | | |
|----------------------------|----------------------|
| 1 Shuttle valve | 4 Air treatment unit |
| 2 Leakage | 5 Belt filter |
| 3 Blow out rest of coolant | 6 Thru-coolant |

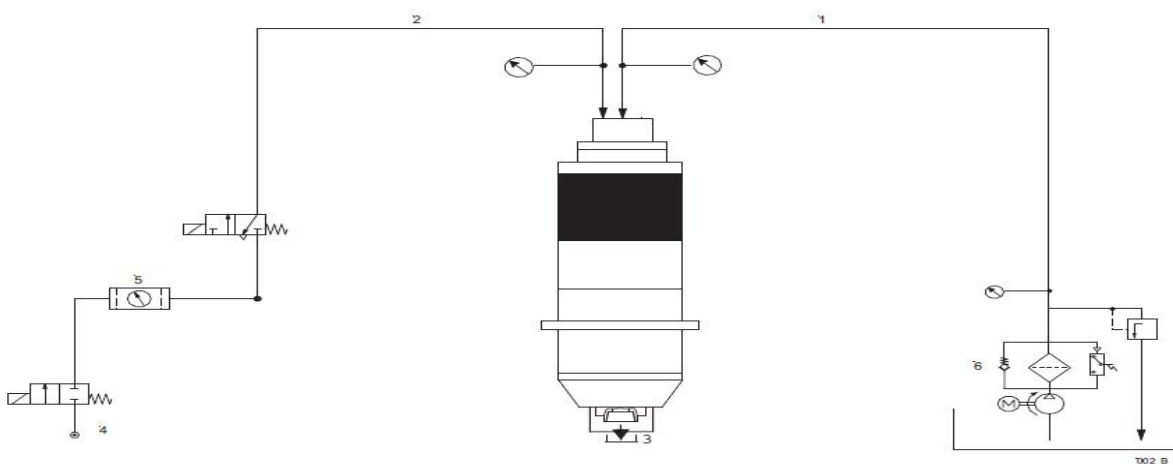


Fig. 4 - 6b: Schematic of MQL System (Minimum Quantity Lubrication)

Legende

- | | |
|-------------------|----------------------|
| 1 Lubricating oil | 4 Air supply |
| 2 Compressed air | 5 Air treatment unit |
| 3 Leakage | 6 MQL unit |

Connection Specifications

For the connection of the thru-coolant system, follow Technical Specifications in Chapter 9.



Information: For more information about the of the thru-coolant system consult the separate operating instructions supplied with the thru-coolant system.

Connection Procedure

Connect the thru-coolant pump unit to the through coolant rotary unit of the spindle according to fig. 4 - 6 and observe the following remarks:

- Observe the installation and service instructions in the documentation of your thru-coolant pump unit.
- Respect the absolute cleanliness during the installation procedure.
- Keep the fittings on the rotary unit closed with blind caps until definitive connection of the piping.
- Install all the periphery units in consideration of the length of lines and prepare the lines accordingly.
- The external diameter of the lines must correspond the line specifications according to chapter 9 "Technical specifications". The internal diameter should be as wide as possible in order to guarantee a maximum flow of air.
- The lines must not be under tension.
- Place the tube connections in a cable duct or secure them to suitable points at the machine.
- The lines must not be bent, respect minimal bend radius during installation of the lines.
- The lines must stand the required pressure and must have the necessary quality.
- Observe the open ends of the drain lines, whose final position must be lower than the position of the through coolant rotary unit.



Caution: The drain lines must be connected always at the lowest point of the through coolant rotary unit and the drain lines must be directed to the bottom. If the thru-coolant can not be drained and it enters the spindle, it will provoke a spindle failure.

4.4.6 Electrical Connections

For the overviews drawings and the cable dimensions for power, PTC, the proximity switches and the encoder, see Technical Specifications, Chapter 9 and Annex, Chapter 10.



CAUTION! Before starting any work on the electrical installation (connection of the spindle to the frequency converter, connection of the proximity switches) ensure that the machine and the frequency converter are disconnected from the mains supply and that the intermediate circuit capacitors (see the operating instructions for the voltage converter) are discharged. **If this is not observed, there is a risk of a dangerous electric shock.**

Connection Procedures

Connect the power and the control cables from the spindle to the frequency converter and to the machine control equipment according to the diagrams in chapter 10 "Annexe" and observe the following points:

- Be sure to observe the installation procedures in the operating instructions for the frequency converter and the machine control equipment.
- Ensure that the connection cables are not under stress.
- If possible, place the connection cables in a cable duct or secure them to suitable points on the machine.
- When connecting the proximity switches to the control circuits, the load must not exceed the following values:
 - Voltage range 10-30 VDC
 - max. load current 200 mA



CAUTION! If these values are exceeded, the inductive proximity switches will be damaged.



Information: Details of the inputs to the frequency converter can be found in the separate operation instructions.

4.4.7 Installing & Connecting Monitoring Devices

For monitoring the operation of the spindle, in addition to the proximity switches and the encoder installed in the spindle, the following devices are required. For specific figures, see Technical Specifications, Chapter 9.

- **Pressure Switch for Oil-Air Lubrication System**

The pressure in the pneumatic system must be continuously monitored. This is achieved by installing a pressure switch before the dosage unit. This switch applies an electrical signal to machine control as soon as the pressure falls below the minimum specified value according to chapter 9 "Technical specifications".



Information: Please consult the separate instructions included with your pressure switch for details on its connection.

- **Spindle Motor Coolant Flowrate Detector**

Motor coolant flow must be continuously monitored by a flow detector to be installed in the return flow of the cooling system. This detector sends immediately an electrical signal to machine control when the coolant flowrate drops.



Information: For details on connecting the flow detector, please consult the separate operating instructions included with your cooling system.

- **Spindle Motor Coolant Temperature Control**

Coolant temperature must be continuously monitored by temperature sensors to be installed at the pump inlet and outlet. They send immediately an electric signal to the machine control when the temperature limit is exceeded.



Information: For details on connecting the temperature sensor, please consult the separate operating instructions included with your cooling unit.

- **Spindle Motor Current Limiter in the Frequency Converter**

Adjust or program the maximum current by means of the current limiter of the frequency converter in order to prevent any damage of the motor.



Information: For details on the current limiter, please consult the separate operating instructions included with your frequency converter.

- **Zero-Speed Control (if any)**

For safe toolchange, the control system must receive a signal indicating a complete real-time spindle stop. The stop signal comes from Proximity Switch S1 or an encoder to be processed next by the zero-speed control and passed on to the machine control system.



Information: Please consult the separate instructions included with your zero-speed control unit for details on its connection.



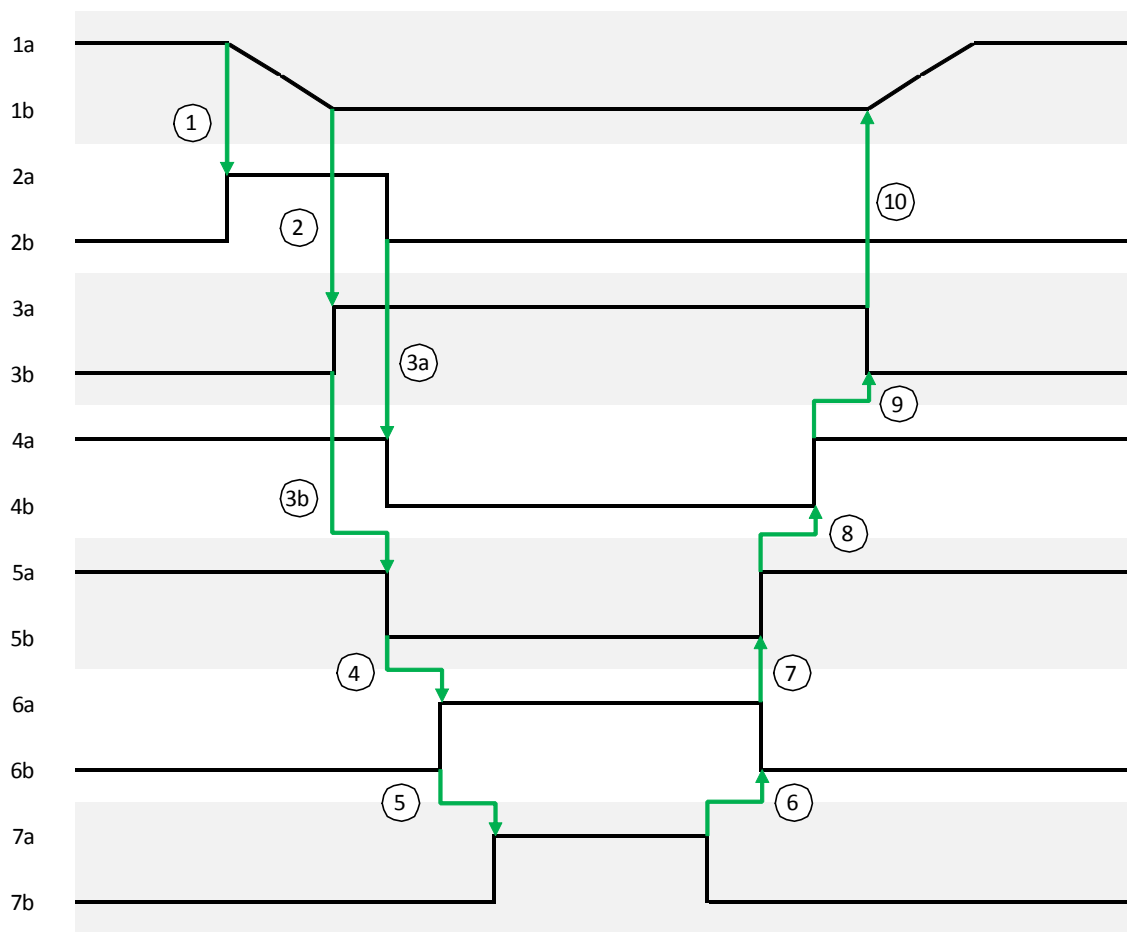
Information: The zero-speed control unit "DA 0025" is available at Fischer AG. This unit meets the specifications required for these functions.

4.5 SPC Info

In order to achieve correct and safe operation of the spindle, the following items must be considered for the design of the Stored Program Control system:

- Tool change and taper blow out air
- Air curtain (protecting air flow)
- Cooling system

Tool Change & Taper Blowout Diagram



FUNCTION / TASK	MONITORING	STATE		STATE	
Speed condition		1a	turns	1b	stands
Blowing out internal cooling pipe		2a	on	2b	out
Spindle standstill monitoring	Sensor S1	3a	attenuated	3b	unattenuated
Tool clamping	Sensor S4	4a	attenuated	4b	unattenuated
Taper blowout		5a	out	5b	on
Tool unclamping (pushed out)	Sensor S2	6a	attenuated	6b	unattenuated
Tool change		7a	blocked	7b	approved

CAUTION: When using KSM (via IKZ), the line must be blown or sucked out before loosening the tool. This prevents the plug, tool holders and tool changer from getting dirty with KSM. In addition, there is a risk that the medium passes through the holes of the cone cleaning in the spindle. After blowing, make sure that the pipe is empty.



CAUTION: The sealing air must be switched on at all times so that the leakage in the rotary feedthrough functions and the KSM does not get into the spindle!



Note: The check valve in the compressed air line should be as close as possible to the spindle. Only then can the most efficient blowing out of the KSM line be achieved.

Air Curtain and Motor Cooling System

In order to prevent the penetration of dirt and tool coolant thorough the protecting air system (labyrinth sealing) to the interior of the spindle, the operating time of the protecting air flow and motor cooling systems must be programmed accordingly. Both systems must be switched on **by the main switch** of the machine **at the same time** and keep on running **continuously**. The sealing air and cooling of the spindle must be maintained for about 10 minutes straight after shutdown (power off) the system. This way it is secured, that no contaminations may reach the inside of the spindle, because of the created vacuum by colling down process.

4.6 Adjusting & Checking Procedures

4.6.1 Adjusting Procedures

To carry out the adjustment procedures, it is assumed that the installation work on the spindle and periphery devices has been completed and that the machine control equipment is functioning correctly.

Pneumatic System

1. Open the mainvalve of the compressed air supply to the air treatment unit.
2. Adjust the **input pressure P1** to the air treatment unit according to chapter 9 "Technical specifications".
3. Open the valve of the protecting air and adjust the **dynamic pressure P2 of the protecting air flow** according to Chapter 9 Technical Specifications at a maximum distance of 15cm from the spindle.



Caution! High thru coolant pressures require setting the dynamic pressure of the protecting air curtain to its upper limit according to Chapter 9 Technical Specifications, in order to prevent thru coolant penetration into the spindle via the air curtain channels.

4. Open the valve of the taper blow out air and activate the clamping system to the position unclamped. Adjust **Dynamic pressure P2 of the taper blowout air** according to Chapter 9 Technical Specifications at a maximum of 15cm from the spindle.
5. Open the valve of the air supply for the dosage unit and adjust the pressure according to Chapter 9 Technical Specifications.
6. Adjust the pressure control device of the dosage unit according Chapter 9 Technical Specifications.



Caution! The dynamic pressures P2 must be observed. After replacing components and changes of pressure within the air supply system, the dynamic pressure P2 must be checked and readjusted 15 cm before the spindle connector. If the pressure is not correctly set the spindle could be damaged and breakes down.

Hydraulic Power Unit

1. For details, see the separate documentation for your hydraulic power unit.
2. Fill the tank of the hydraulic power unit with the specified **hydraulic oil** (see chapter 9.3).
3. The line to the spindle must be absolutely exhausted..
4. Adjust the pressure according to Chapter 9 “Technical data”

Cooling System

1. For details, see the separate documentation for your cooling system.
2. Check coolant flowrate, see Chapter 9 “Technical data”
3. Adjust the **thermostat for the recooling** to a temperature between **20° and 25 °C**. (see the documentation for the cooling system).
4. Adjust the **flow detector, or pressure detector** to a value **1 l/min, or 1 bar below the actual flow**.
5. If your cooling system is also equipped with a heater, set the **heating thermostat** to **20 °C**. (see the documentation for the cooling system).

Minimum Quantity Lubrication System / Thru-Coolant System for Spindles with Rotary Unit

1. For details, see the separate documentation for your minimum quantity lubrication system / thru-coolant system.
2. Adjust the pressures according to chapter 9 “Technical specifications”

Frequency Converter

Adjust the operating parameters as listed in the documentation for the frequency converter, while still observing the spindle motor data.



ATTENTION! For synchronous motors, see also Para. 4.7.1: Entering Commutation Angle Offset for Synchronous Motors.

Oil-Air Lubrication System



WARNING! Ensure absolute cleanliness when working on the oil-air lubrications system.

1. Fill the tank of the oil-air lubrication unit with the **specified lubrication oil** (see chapter 9 "Technical specifications").
2. Remove the plug from the free end of the lubrication oil tube (connection between dosage unit and spindle) and place the open tube into a collecting vessel.
3. Switch on the oil-air lubrication unit.
4. On the oil-air lubrication control unit adjust the interval times or cycles according to chapter 9 "Technical specifications" or the "Operating Instructions" of the oil-air lubrication unit.
5. Bleed the dosage unit. Open the vent screw until bubble-free oil is discharged and then close the vent screw again (see the operating instructions for the oil-air lubrication unit).
6. Then let the oil-air lubrication unit run for **4 - 5 hours**. This will ensure that any impurities/slush oil in the dosage unit are washed out of the lines.
7. Remove the locking plugs from the lubrication oil connections on the spindle and connect the lubrication oil lines (Legris or Eisle connectors.)
8. Set **Dynamic pressure P2** (for figures, see Chapter 9: Technical Specifications) on each individual lubrication connection at a maximum distance of 15cm.



Information! After the dynamic pressure on the lubrication connections has been set the manometer used for the measurement must be removed.

4.6.2 Checking Procedures

Before the first time operation of the spindle check and make sure that:

- all securing screws on the spindle are properly tightened.
- all tubes are correctly connected, suitably labelled, securely fixed, properly sealed and not buckled.
- all cables, especially those of the PTC temperature sensors, are correctly connected acc. to Chapter 10: Annex.
- the pressures in the oil-air lubrication , air curtain, taper blow out, hydraulic clamping and cooling systems have been correctly set acc. to Chapter 9: Technical Specifications.
- all pressure sensors, flow meters and thermostats (pneumatic and cooling systems) function correctly.
- the parameters set on the frequency converter match the parameters of the motor acc. to Chapter 10: Annex.
- the interval time set on the oil-air lubrication unit is permitted for your spindle acc. to Chapter 9: Technical Specifications.
- the synchronous motors have a correct Commutation Angle Offset entered. See Para. 4.7.1: Entering Commutation Angle Offset for Synchronous Motors.
- the tanks of the oil-air lubrication unit, the cooling system and the hydraulic power unit have been filled with the correct material acc. to Chapter 9: Technical Specifications
- the levels in the tanks of the oil-air lubrication, cooling and hydraulic systems are correct (see the operating instructions for the peripheries concerned)

4.7 First-Time Operation



WARNING! Before putting the spindle into operation for the first time, all installation and adjustment procedures must have been completed and the system (spindle and periphery devices) checked.

For the first-time operation of the spindle carry out the following procedure:

1. Turn the main switch of the machine on.
2. Switch all periphery devices on.
3. **Check** every transparent oil-air lubrication line with the aid of a light source (torch, flash light) whether **oil is flowing** regularly towards the spindle.
4. Place a test taper mount or a dynamically balanced tool holder in the spindle and clamp it.



CAUTION! Never run the spindle unless it has a tool or a test taper fitted and clamped. If this is ignored, the interface (especially the HSK interface) will be damaged.

5. **Wait approx. 10 to 15 minutes** after the periphery devices have been switched on and check the oil which must return visibly through the oil return lines / holes.
6. Switch on the spindle and keep it running for **approx. 5 minutes at 20 % of the maximum speed**.
7. Increase the spindle speed in 10 % increments. Hold for 5. min. each, the speed is constant up to the maximum speed achieved.
8. Keep the spindle running at maximum speed for approx. half an hour.
9. During this period check the spindle temperature and vibration according to the test protocol. If the temperature is too high, the spindle must be switched off and the cooling system checked.
10. Insert all toolholders one after the other and let the spindle run at the maximum speed for each tool (without exceeding the maximum speed of the spindle) and check how the spindle operates.



CAUTION! Ensure that only toolholders/tools are inserted which meet the specifications described in chapter 3.5.



CAUTION! The vibrations should not be significantly greater than those produced when a test taper or a well balanced tool holder was fitted. If stronger vibrations are produced, check the tool or tool holder.

4.7.1 Entering Commutation Angle Offset for Synchronous Motors



ATTENTION! Whenever you change the spindle, change also the Commutation Angle Offset in the drive parameters of the frequency converter. Each spindle has its own Commutation Angle Offset recorded in a Test Sheet to be found with each spindle delivery. Check and correct as required the Offset on any mounting / dismounting of the spindle, ie. exchanges, repairs, revisions, checks.

5. Operation

5.1 Safety Measures During Operation



WARNING! The following guidelines on safety and hazards must be strictly observed. Should you ignore them, a danger to persons and/or damage to the spindle or other property may ensue.

- The spindles must be operated by skilled personnel (e.g. machine operator). The **customer is responsible** for assigning the personnel and ensuring that they are qualified and competent to operate the spindle.
- Unless it can be safely operated, switch the spindle off and ensure it against random activation.
Safe operation of the spindle is no longer possible when:
 - the spindle is visibly damaged,
 - the spindle operates abnormally: strong vibrations, loud knocking sound, etc.
 - one or more peripherals fail

If in doubt, please contact After Sales Service of Fischer AG.

- Generally applicable instructions and guidelines for safe working condition must be strictly observed.
- **Only fully tested tools are permitted for use** (centrifugal force test according to EN ISO 15641, dynamically balanced to quality rating G2.5 according to VDI 2060 or ISO 1940) which have the **maximum rotation speed engraved** on them. The **maximum speed must not be exceeded**.
- **Safety equipment fitted must not be removed or bridged**. Operation of the spindle with open protection cabin is forbidden.
- Without a written permission of Fischer AG, **absolutely no modifications or extensions** to the spindle or its peripherals must be made.

5.2 Starting Up and Running Down the Spindle

Daily Start Up Procedure

To start up the spindle carry out the following procedure:

1. Turn on the main switch of the machine.
2. After switching on, make the following checks on the machine:
 - Check that all peripheral to the spindle are operating and that their operating parameters are within the specified tolerances.
 - Check visually the flow of oil inside the lubrication tubes by using a light source. A fine regular flow of oil towards the spindle must be visible in all the tubes.



CAUTION! If any of these checks prove negative, do not operate the spindle. Report the situation to the person responsible for maintenance.

3. Clean the taper mount (collet) and the tool holder.



CAUTION! Do not put your fingers in the taper mount of the spindle. They could be crushed.

4. Insert the tool and clamp it.



CAUTION! Never operate the spindle without a fitted and clamped tool.

5. Start-up and run-up the spindle:



CAUTION! Never run-up the spindle to maximum speed during the start-up time when it is cold, otherwise, due to different thermal expansion characteristics in the bearings, stress conditions are produced, which can cause damage to the spindle.

Daily Run Down Procedure

To put the spindle out of operation carry out the following procedure:

1. Set the spindle to zero speed.
2. Remove the tool/ toolholder from the taper mount (Steep tapers).
3. Tool/ toolholder can be clamped (HSK- tapers).
4. Turn the main switch of the machine off.
5. Put a taper cleaner in the taper mount.

5.3 Attending the Spindle in Operation

The attendance is limited to checking if the spindle operates correctly and if the conditions for toolchange are met.

Checking Spindle Operation

The operation of the spindle should be continuously monitored. If any changes are detected which could impair safe operation, put the spindle out of operation immediately and ensure it against random activation.

Safe operation of the spindle is no longer possible when:

- the spindle has visible damage,
- strong vibrations are detected,
- unusual noises are audible,
- one or more of the peripherals fails.

Toolchange Conditions

For each and every toolchange:

- the spindle must be at a complete stop.
- The taper in the spindle and on the tool must be **clean and without damage**.
- Before use, all tools must be inspected. Only tools meeting the specifications set out in Para. 3.5 must be used.



CAUTION!

For manual toolchange, always protect your hands with leather gloves.

6. Maintenance

6.1 Safety Precautions for Maintenance Works



WARNING! The following guidelines on safety and hazards are to be strictly observed. If this warning is ignored, it could result in danger to persons and/or damage to the spindle or other property.

- All maintenance work is to be carried out by fully trained skilled personnel. The customer is responsible for ensuring that personnel is qualified to carry out the various tasks.
- Personnel who will have the task of carrying out maintenance work on periphery devices must have read and understood the relevant documentation before starting this work.
- Individual maintenance tasks have to be carried out when the machine or periphery devices are switched on. For this reason the instructions for the maintenance work must be strictly observed and implemented.
- Before starting any maintenance work on pneumatic and hydraulic systems, these units must be made completely safe (isolate the units from the mains, no pressure in the systems, etc.)

6.2 Maintenance Schedule

We assume, that the average of the machine operators are producing in a two shift interval with their High-Speed-Cutting Centers, i.e. the operating hours are based on 15 hours per day, 5 days per week or 48 weeks per year.



ATTENTION: If the machine operators work on a one, three or four shift basis, the maintenance schedule must be extended or shortened accordingly.

Daily Maintenance

What	Who
Check the oil flow in the lubrication lines (dosing unit - spindle).	machine operator

Weekly Maintenance

What	Who
Clean and check tools, tool holders and tool taper mount	machine operator
Check oil level in oil-air lubrication unit and refill if necessary	machine operator
Empty oil separator	machine operator

Monthly Maintenance or After 300 Hours' Operation

What	Who
Check oil level in hydraulic unit and coolant level in cooling unit and refill if necessary.	machine operator

Semi-Annual Maintenance or After 1800 Hours' Operation

What	Who
Replace fine filter element in air treatment unit	skilled worker
Test clamping force of clamping system and if necessary regrease clamping device	skilled worker

Annual Maintenance or After 3600 Hours' Operation

What	Who
Replace filter element in oil separator	skilled worker
Check and replace hydraulic oil (if necessary)	skilled worker
Check and replace coolant (if necessary)	skilled worker

6.3 Maintenance Procedures



CAUTION! The oil-air lubrication system and protecting air must be switched on when maintenance work is carried out on the spindle.

What	When	Who
Check oil flow in lubrication lines near the spindle connector .	every day	machine operator

Proceed as follows:

1. Switch on the oil-air lubrication system.
2. Inspect visually each lubrication tube between dosing unit and spindle. A fine regular oil flow must be visible in all lines. Use a light source to illuminate the tubes. If this not the case, do not run the spindle or switch it off immediately if it is already running. Report the situation to the person responsible for maintenance.

What	When	Who
Clean and regrease HSK taper and tool holders	every week	machine operator

Proceed as follows:

1. Remove the tool/tool holder from the spindle.
2. Switch the **machine off** (main switch to OFF and secured in this position).
3. Clean the HSK taper of the spindle using a taper cleaner which should be inserted in the taper and rotated 2 to 3 times and all the tool holders.
4. Clean and regrease the tool holders specially the interiour, where the clamping device clamps the tool holder. Use grease according to chapter 9 "Technical specifications".

What	When	Who
Check oil level in oil-air lubrication unit and refill if necessary.	every week	machine operator

Proceed as follows:

1. Check the oil level in the oil-air lubrication unit.
2. Refill with oil if the level is within 1 cm of the minimum mark. Top up to approx. 1 cm below the maximum mark.



CAUTION! The oil used for refilling must meet the quality standards specified in chapter 9 "Technical Specifications" or chapter 10 "Annex". If this is not observed, the life time of the bearings will be considerably reduced.

CAUTION! Observe and implement the directions indicated in the documentation for your oil-air lubrication unit.

What	When	Who
Empty oil separator	every week	machine operator

Proceed as follows:

1. Switch the spindle off (main switch of the machine set to OFF and secured in this position).
2. Disassemble the oil separator as described in the instructions.
3. Dispose of the oil.



CAUTION! Never use oil a second time. Oil can pollute the environment. Ensure that the oil is correctly disposed of. Observe any local regulations and laws regarding the disposal of oil.

What	When	Who
Check oil level of hydraulic unit and refill if necessary.	every month	machine operator

Proceed as follows:

1. Check the oil level in the hydraulic power unit.
2. Refill with oil if the level is below the minimum mark (inspection glass). Top up to the maximum mark.



CAUTION! The oil used for refilling must meet the quality standards specified in chapter 9 "Technical Specifications" or chapter 10 "Annex".

CAUTION! Observe and implement the directions indicated in the documentation for hydraulic power unit.

What	When	Who
Replace fine filter element in air-treatment unit	every 6 months	skilled worker

Proceed as follows:



CAUTION! When working on the pneumatic system, ensure absolute cleanliness. Dirt in the pneumatic system can create problems when operating the spindle.

1. Switch the spindle off (main switch of machine to OFF and secured in this position).
2. Separate the spindle from the compressed air network.
3. Wait until the pressure indicated on the manometer has fallen to 0 bar.
4. Replace the filter according to the procedure in the instructions for the air treatment unit.

What	When	Who
Check clamping force of clamping system and regrease clamping device if necessary	every 6 months	skilled worker

Proceed as follows:

1. Switch off the spindle (main switch of the machine set to OFF and secured in this position).
2. Insert the clamping force test unit. (Ott-Jakob Power Check) and measure the clamping force. If the measured value is below the relevant figure in Technical Specifications, Chapter 9, the clamping device must be disassembled, cleaned, inspected, regreased, reassembled and readjusted. If a clamping force test unit is not available or nothing changes the spindle has to be returned to the company Fischer AG for further testing and repair (Belleville springs, clamping system).



WARNING! A spindle which has a clamping force below the minimum value according to Technical Specifications, Chapter 9 must not be put into operation because safe clamping of the tool can no longer be guaranteed. **If this warning is ignored, persons could be injured or killed and/or property damaged.**

What	When	Who
Replace filter element in oil separator	every year	skilled worker

Proceed as follows:

1. Switch the spindle off (main switch of the machine set to OFF and secured in this position).
2. Replace the filter according to the procedure set out in the instructions for the oil separator.

What	When	Who
Replace hydraulic oil	every year	skilled worker

Proceed as follows:

1. Switch the spindle off (main switch of the machine set to OFF and secured in this position).
2. Replace the oil according to the procedure set out in the documentation for the hydraulic power unit.



CAUTION! The oil used for refilling must meet the quality standards

What	When	Who
Replace motor coolant	every year	skilled worker

Proceed as follows:

1. Switch the spindle off (main switch of the machine set to OFF and secured in this position).
2. Replace the coolant according to the procedure set out in the documentation for the cooling unit.



CAUTION! The motor coolant must meet the quality standards given in Chapter 9 "Technical Specifications" and/or Chapter 10 "Annex".

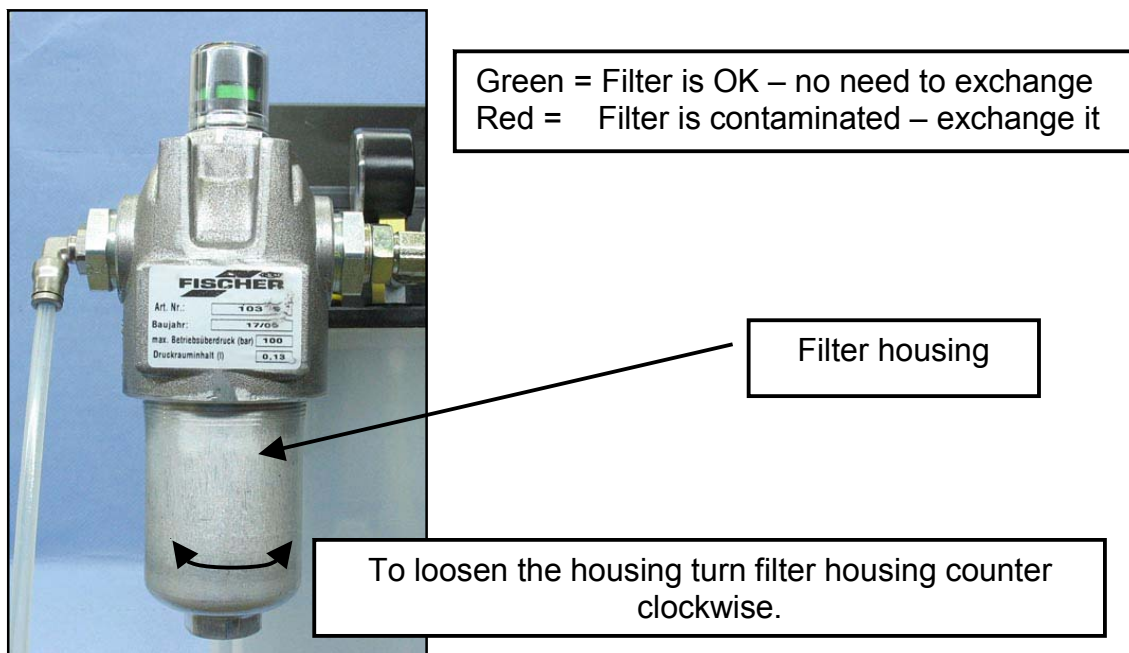
6.3.1 Replacing Oil Filter Element

If the filter is contaminated and the required flow of oil is not longer ensured it will be indicated by the indicator device on top of the filter. If the filter element is not exchanged when contaminated it will cause a reduced oil flow to the dosing unit and it is not ensured that the spindle bearings will be provided with the appropriate amount of oil. Bearing failure will be the most likely result.

It is required that the compressed air supply to the dosing unit and the labyrinth sealing air is switched off.

Any work has to be carried out under clean surrounding. If contamination enters into the line between the oil filter and the dosing unit it will have direct negative influence to the life time of the spindle.

Make sure that the ambient surrounding is clean. The oil filter shall be cleaned externally prior to the exchange process of the oil filter. If there is contamination on the oil filter or in the direct surrounding area there is a risk that this will enter the interior when removing the filter housing.



To replace the filter element, proceed as follows:

1. Stop spindle.
2. Switch off air supplied to the dosing unit.
3. Switch off air supplied to the labyrinth seal air connection.
4. Disconnect the oil-air unit from the power and make sure that the oil hose from the filter to the dosing unit is not pressurized.
5. Remove the oil air tubes from the spindle connectors and lead them into a clean container.
Pay attention that they must not get contaminated.
6. Close the connections of the spindles with the shipping plugs in order to ensure that no contamination enters the connectors till the tubes are not connected.
7. To exchange the oil filter unscrew the filter housing counter clockwise by using an open jaw wrench size 24 mm
8. Remove the filter element.
9. Insert new filter element.
10. Screw the filter housing clockwise till it is tightened.
11. Bleed the oil system as described in the Assembly Instructions of the spindle.
12. Flush the oil-air tubes for 4-6 hours.
13. Connect the oil-air tubes to the spindle connector and adjust the oil-air pressure at the dosing unit according to the spindle Assembly Instruction.
14. After the pre-lubrication cycle is accomplished the spindle needs a new installation warm-up – run in procedure.

6.4 Protect Environment



WARNING! Fuels (oils, coolants, etc.) can cause environmental pollution. Therefore handle such materials with care and ensure that used oils, coolants, etc. are correctly disposed of or recycled.

In addition all local regulations and laws on handling fuels and on protecting the environment must be observed.

7. Troubleshooting & Repairs

7.1 Safety Precautions for Troubleshooting & Repairs



WARNING! The following guidelines for safety and hazards are to be strictly observed and implemented. If this warning is ignored, it could result in danger to persons and/or damage to the spindle or other property.

- All work associated with trouble shooting and repair is only to be carried out by fully trained skilled personnel. The customer is responsible for ensuring that personnel is qualified to carry out the work.
- Before starting any worked associated with trouble shooting and repair, carry out the switching off procedure described in paragraph 5.2 and ensure that the machine and periphery devices cannot be inadvertently switched back on again.
- Should, for any reason, work need to be carried out when the machine is running, this work is to be assigned to a skilled person who is full aware of the danger involved. In addition a suitably instructed assistant should be present who can switch the machine off in an emergency.
- When spindle components need to be replaced, use exclusively original spare parts supplied by the company Fischer AG.

7.2 Troubles & Remedies

Trouble	Cause	Check / Remedy
Tool not correctly clamped.	Metal chips, too much grease or dirt restrict the clamping process. Dimensional accuracy of tool holders out of tolerance. Position of tool is not sufficiently lined up with tool changer. Adjustment of clamping cone not correctly set, lock screw loose. Clamping unit worn or damaged Cup springs broken, clamping force insufficient or any other defect.	Clean and check HSK/ISO Interface. Check proximity switch S4 with Ott-Jakob POWER CHECK. Check, exchange or replace faulty tool holder. Line up tool changer with the spindle ± 0.1 mm (see operating instructions of machine). Readjust to correct setting, see section 7.3.1. Replace clamping unit Return spindle for repair to Fischer AG
Tool cannot be released.	Tool holder got stuck. Toolholder chuck dimension is out of tolerance or damaged. Hydraulic cylinder defective. Oil pressure (tool unclamping) too low. Frictional corrosion (rust) on taper in consequence of extreme vibrations. Oil level too low.	Increase pressure in hydraulic power unit by 20 bar. Afterwards reduce it to the normal pressure according to chapter 9 "Technical specifications" (see operating instructions for hydraulic power unit). Check the toolholder chuck and replace if required. Check hydraulic power unit or set hydraulic oil pressure according to chapter 9 "Technical specifications" (see operating instructions for hydraulic power unit). Increase pressure in hydraulic power unit by 20 bar. Afterwards reduce it to the normal pressure according to chapter 9 "Technical specifications" (see operating instructions for hydraulic power unit). Check quality of dynamical balance of the tools and toolholders, reduce advance speed Top up oil.

Trouble	Cause	Check / Remedy
Tool released during working cycle.	Gripper segments broken. Clamping taper or draw bar broken. Belleville springs broken. Pull-in force insufficient.	Replace gripper segments. Return spindle for repair to Fischer AG. Return spindle for repair to Fischer AG. Check reference dimension. If dimension is correct return spindle for repair to Fischer AG.
Spindle vibrates.	Tool or tool holder not correctly balanced. Concentricity of tool out of tolerance. Normal or abnormal wear of bearings.	Set dynamic balance of tool and tool holder to quality rate G2.5 (see standards VDI 2060 or ISO 1940). Maximum concentricity tolerance, see test sheet. Return spindle for repair to Fischer AG
Proximity switch S2, tool unclamped, does not function.	Fault in the electrical connection. Clamping unit (clamping cone) is not set correctly. Proximity switch is not set correctly. Proximity switch is defective. Clamping system is defective.	Check connections and connecting cable Check reference dimension of the clamping cone and readjust if necessary. Check reference dimension of the proximity switch and readjust if necessary. * Replace proximity switch. * Check clamping system and replace if necessary. * If the fault can not be found, return the spindle for repair to Fischer AG.



Note: * Asterisked repair works must be carried out by qualified Fischer AG-trained personnel. Otherwise send the spindle to Fischer AG or request a service technician.

Trouble	Cause	Check / Remedy
Proximity switch S3, tool clamped without tool, does not function.	Draw bar is bent by means of a crash.	Check reference dimension and condition of the draw bar, if necessary replace clamping system.*
	Fault in the electrical connection.	Check connections and connecting cable.
	Proximity switch is not set correctly.	Check reference dimension of the proximity switch and readjust if necessary.*
	Proximity switch is defective. Clamping system is defective.	Replace proximity switch * Check clamping system and replace if necessary.* If the fault can not be found and repaired, return the spindle for repair to Fischer AG.
Proximity switch S4, tool clamped, does not function.	Metal chips, too much grease or dirt restrict the clamping process.	Clean HSK/ISO interface and toolholders. Disassemble, check, regrease and reassemble clamping unit. Check tool holders.
	Fault in the electrical connection.	Check connections and connecting cable.
	Clamping unit (clamping cone) is not set correctly.	Check reference dimension of the clamping cone and readjust if necessary.
	Proximity switch is not set correctly.	Check reference dimension of the proximity switch and readjust if necessary.*
	Proximity switch is defective. Clamping system is defective.	Replace proximity switch.* Check clamping system and replace if necessary.* If the fault can not be found, return the spindle for repair to Fischer AG.



Note: * Asterisked repair works must be carried out by qualified Fischer AG-trained personnel. Otherwise send the spindle to Fischer AG or request a service technician.

Trouble	Cause	Check / Remedy
Spindle does not rotate.	Defective encoder/zero speed control.	Check signals of encoder/zero speed control and replace if necessary.
	Electrical connection interrupted.	Check electrical connections and replace if necessary.
	Defective frequency converter	Check the voltage at the output of the frequency converter. If no voltage appears, consult service instructions of frequency converter.
	Shorted turns.	Measure the difference in resistance of the windings. The difference in resistance of the individual motor phases must not exceed 0.1 Ohm. If the difference exceeds this figure, the spindle will have to be returned for repair to Fischer AG.
	Short circuit to earth.	Check all electrical connection for short circuit to earth. If a short circuit is still present when the mains plug is pulled out, the spindle will have to be returned for repair to Fischer AG.
Temperature sensor in stator windings indicates error.	Cooling system not switched on.	Switch cooling system on.
	Faulty electrical connection to PTC.	Measure resistance between frequency converter and spindle and if necessary replace electrical connection.
	Spindle overloaded.	Reduce load.
	Temperature sensor defective.	Measure resistance of sensor. If resistance is above 3000 Ohm at room temperature, the spindle will have to be returned for repair to Fischer AG.

Trouble	Cause	Check / Remedy
Proximity Switch S1 (if any) , zero speed control, does not function.	Fault in the electrical connection. Proximity switch is not set correctly. Proximity switch is defective.	Check connections and connecting cable. Check reference dimension of the proximity switch and readjust if necessary * Replace proximity switch * If the fault can not be found, return the spindle for repair to Fischer AG
Proximity Switch S5 (if any) , piston cylinder, does not function.	Piston of the cylinder isn't in end position (tool clamped). Fault in the electrical connection. Proximity switch is not set correctly. Proximity switch is defective.	Check discharge (cylinder) tool clamped. Check electrical connection. Check reference dimension of the proximity switch and readjust if necessary.* Replace proximity switch.* If the fault cannot be found, return the spindle for repair to Fischer AG.



Note: * Asterisked repair works must be carried out by qualified Fischer AG-trained personnel. Otherwise send the spindle to Fischer AG or request a service technician.

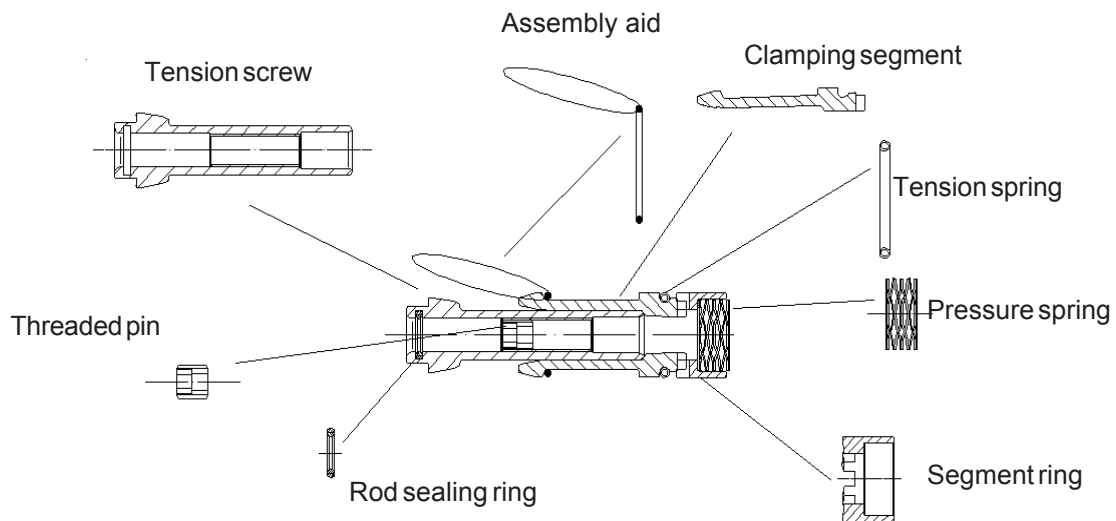
7.3 Repairs



WARNING! The following guidelines for safety and hazards are to be strictly observed and implemented. If this warning is ignored, it could result in danger to persons and/or damage to the spindle or other property.

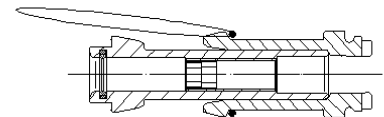
CAUTION! With the exception of the following repair work, the user is not permitted to carry out any other work on spindles.

7.3.1 Gripper Components

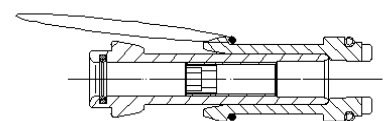


Assembly sequence for the gripper components

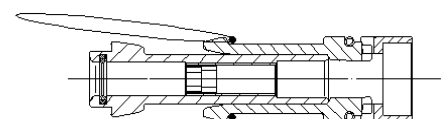
- Lubricate degreased components with any oil for refitting.
- Fit assembly aid to tension screw. Fit clamping segments below the O ring of assembly aid.



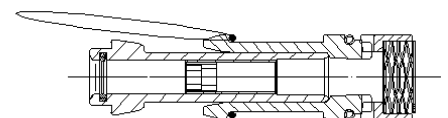
- Fit tension spring and adjust the butt-joint of the spring centrally to one of the clamping segments.



- Arrange clamping segments radially and uniformly distributed. Fit segment ring, here the noses of the segment ring must engage in the cut-outs in the clamping segments.

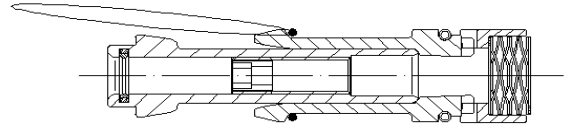


- Push pressure spring into the segment ring. When required the pressure spring can be fixed in position with any grease. Gripper is ready to fit.

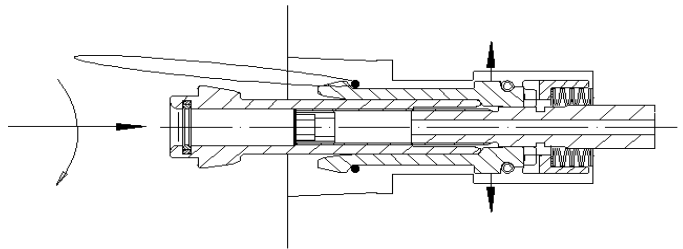


Installation sequence, spindle assembly

- Gripper ready to fit.

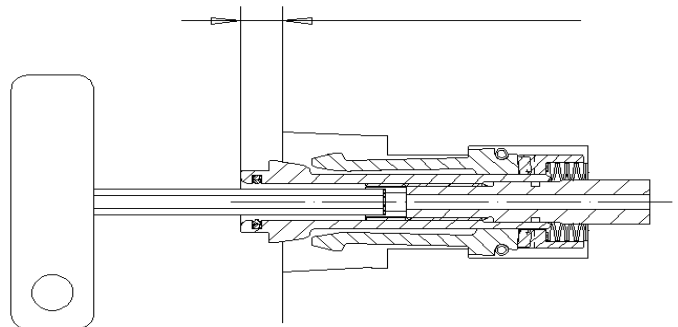


- Bring the tensioning rod into the tool-ejection position. Enter gripper into the spindle and screw onto the tensioning rod to cause the clamping segments to move into their working position. This movement can be heard as a "Clicking" sound. Remove fitting aid.



Reference Dimension

- Screw in the gripper deeper. Set reference dimension with approx. -0.1. Hold tension screw with an open end wrench and lock the threaded pin over the hexagon screw. Check and, if necessary, reference setting dimension.



Attention!

Check the reference dimension prior to the first use of a tool!

Dismantling:

Release threaded pin and tension screw and remove gripper components from the spindle.

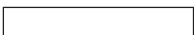
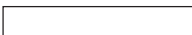
7.3.2 Clamping Force Measuring Device POWER-CHECK from Ott-Jakob



Note! The clamping force measuring device "Power check" is used for checking the clamping force periodically during maintenance and for checking the setting of the proximity switches, especially after replacing the clamping unit. The use of the "Power check" device can be read in the appropriate operating instructions. The setting of the "Power check" can be found in Chapter 7.3.4 below.

7.3.3 Checking Proximity Switch Settings for HSK-Interface Spindles

CHECKING DIAGRAM

<i>Tool unclamped</i>		<i>Tool clamped</i>		<i>Clamped without tool</i>
		Power-Check Setting (Ott-Jakob) - 0.25	Power- Check Setting (Ott-Jakob) - 0.15	
S2				
S3				
S4				
S5 *				



Proximity switch active (Signal lights up)



Proximity switch inactive



Proximity switch active / inactive acc. to HSK tolerances

Proximity Switches

S2 Tool unclamped (must be activated, when tool is completely unclamped, i.e. 0.4 - 0.6 mm in front of the final position, **measured during unclamping**.)

S3 Clamped without tool (active between 0.3 - 0.5 mm in front of the final position of clamping cone, **measured during clamping**.)

S4 Clamped with tool.

S5 (* if any) Monitoring piston stroke limit.



Note: The proximity switches have to be checked in cold and warm spindle condition. The operating temperature can be reached by starting up and stopping the spindle several times.

7.4 Customer Services

For repair work which cannot be carried out by the customer, proceed as follows:

- Disassemble the spindle described in paragraph 8 and pack it according to instructions (wrapped in anti-corrosive paper) in its original wooden case.
- Make a copy of the repair formula in chapter 10 "Annex" and fill in the following information:
 - Full address (including the name of the person to contact if any questions arise)
 - Spindle number
 - A precise description of the defect (e.g. proximity switch defective, clamping system defective, etc.)
 - Date and signature
- Put the repair form in the wooden box with the spindle and send it to the company Fischer AG (the precise address can be found on the repair form).

By carrying out this procedure you will make your contribution to a quick repair.

8. Decommissioning

8.1 Spindle Decommissioning Safety



WARNING! The following guidelines for safety and hazards are to be strictly observed and implemented. If this warning is ignored, it could result in danger to persons and/or damage to the spindle or other property.

- All work associated with decommissioning the spindle is only to be carried out by fully trained skilled personnel. The customer is responsible for ensuring that personnel is qualified for the various tasks.
- When starting the decommissioning work, the machine and individual systems will still be connected to the mains voltage supply and some systems will still be under pressure. Therefore be sure to observe the instruction for running down the spindle.
- Be sure to observe local instructions and regulations on the handling of fuels and oils (oil, motor coolant, etc.).

8.2 Decommissioning & Disassembly

Decommissioning Procedures

To decommission the spindle carry out the following procedure:

1. Switch the spindle off and ensure it is not rotating.
2. Remove the tool from the spindle.
3. If available switch the hydraulic power unit or the pneumatics system off. Release the pressure from your tool clamping system.
4. Interrupt the compressed air supply and wait until the pressure in the system has fallen.
5. Disconnect the machine from the mains power supply and ensure that it cannot be switched back on inadvertently (**main switch to OFF** and secured in this position).



WARNING! Before continuing with this work wait until the intermediate circuit capacitors in the frequency converter have completely discharged (see the documentation for the frequency converter).

6. Ensure that the power supply to the periphery devices (hydraulic power unit, oil-air lubrication unit, etc.) has been switched off (check with a voltmeter).

Disassembly

1. Remove the tube connections from the spindle to the periphery devices and mark them if this has not been done already.



CAUTION! Ensure absolute cleanliness. Immediately after removing the tubes seal them and the connections on the spindle with suitable plugs.

2. Disconnect the electrical cable.
3. Aspirez l'eau restante du circuit de refroidissement.
4. Disassemble the spindle. Carry out the procedure described in paragraph 4.3 but in the reverse order.
5. Place the spindle in its original wooden box.

8.3 Storage

Preparing a Used Spindle for Storage

For a used spindle to be kept in storage, proceed as follows:

1. Smear some oil on the spindle sleeve and then pack the spindle in anti-corrosion paper.
2. Place the spindle in its original wooden box which provides the best possible protection against damage and corrosion.
3. Fix a label to the wooden box indicating the **spindle type** and the **date it was put into storage**.

Preparing a New Spindle for Storage

For a new spindle to be kept in storage, proceed as follows:

1. Place the spindle in its original wooden box which provides the best possible protection against damage and corrosion.
2. Fix a label to the wooden box indicating the **spindle type** and the **date it was put into storage**.

Info for Storing Oil-Air & Grease-Lubricated Spindles

Observe the following points for storing the spindles:

- During the storage no aggressive media may influence e.g. gases, nebulas or aerosols of acids, caustic solutions or salts.
- Direct sunlight must be avoided, since it can lead to large variations in temperature in the packing.
- In order to prevent the condensation in the interior of the spindle the temperature day/night changes must be $< 8^{\circ}\text{C}$.
- The permissible storage temperature must be between $+10^{\circ}\text{C}$ and $+40^{\circ}\text{C}$.
- The storage location must be dry and well ventilated.
- The relative humidity of the storage location must be max. 55%.
- No storage outdoors is allowed.



CAUTION! The bearings of oil-air lubricated spindles must be checked after 3 years' storage. The bearing grease for grease-lubricated spindles must be replaced after 2 years' storage.



Information: Should the info points above be disregarded, the spindle operation readiness is no longer guaranteed and the maximum lifetime of the bearings can be seriously reduced.

8.4 Disposal

Should it be necessary to dispose of the spindle, return it to Fischer AG in its original packing. Fischer AG will ensure that the spindle is disposed of correctly with no hazard to the environment.

Fischer AG rejects any liability for whatever results of inappropriate disposal by the customer.

9. Technical Specifications

9.1 Spindle

General

Type	MFW-1920/30/1 VC HSK-A63
Tool interface	HSK-A63
Spindle Article No	114487
Spindle Dwg No	012-001
Fitting set	116362 / 133291
Coolant through-unit	018-004
Lubrication	oil-air DLS
Motor	asynchronous
Coolant motor and bearings	waterbased fluid
Clamping system	mechanically/hydraulically driven
Tool control	by proximity switches
Rotating direction	right/left (cw / ccw)
Positioning	sin/cos 1Vpp
Mounting position	any
Bearing type	ceramic-hybrid
Number bearings	4

Dimensions

Spindle dimensions	see spindle dwg, chapter 10
Weight	70 kg +/-5%

Toolholding

Type	HSK-A63
Clamping force ex works min	22 kN
Release cylinder	Typ Fischer
Hydraulic pressure for tool unclamping	min. 50 bar, max. 60 bar No pressure when spindle is turning!!!
Clamping Set Order No	117089
Forkspanner size	24 mm
Allen key size	8 mm
Reference Dimension X	10.4 - 10.6 mm

Coolant trough-unit

Manufacturer	Fischer AG
Article No	120144
Type	1-K F axial
Coolants	- KSS - MQL (Lubricants < 50 ml/h) - Aerosol (Lubricants > 50 ml/h) - Air
Operation without coolant	Yes
Maximal rotation speed	30'000 min ⁻¹ (maximal spindle rotating speed)
Leakage @ 40 bar	< 1 dl/min.
Leakage @ 80 bar	< 1.5 dl/min.
Air curtain consumption	max. 90 l/min.



Caution

Do never use the trough-rotary-union without air curtain!

Spindle Control

Rotating speed	110055 sin/cos 1Vpp, Ref. z=256, m=0.3
Tool unclamped	Prox. switch S2
Clamped without tool	Prox. switch S3
Tool clamped	Prox. switch S4
Monitoring piston stroke limit	Prox. switch S5
Stator temperature	Heat detector KTY
Bearing temperature	2 Heat detectors PT 100

Electrical Connections / Schematics

Motor	102469
Proximity switches	92000044 + PT100
Encoder	91170-183 + KTY
Option DMD/SGS Mini	92000066
Option VBS 121041	92000007 / 104386
Option VBS 121237	106386 / 104386

Motor Specification

See Chapter 10, Paragraph 10.6.7

Spindle Fittings

Coolant IN/OUT, motor and bearings	CIMB/COMB	Ø 7 mm	
Air curtain for spindle	AC	Ø 8 mm	Legris
Oil-air lubrication	MV	Ø 4 mm	Legris
Oil return	OR	Ø 8 mm	Legris
Tool unclamping	TU	Ø 8 mm	Ermeto
Taper blowout	TB	Ø 8 mm	Legris
Coolant through-unit, spindle	DDF	Ø 8 mm	Ermeto
Cooling lubricant, external	KSME	Ø 7 mm	
Air curtain for coolant through-unit	ACD	Ø 6 mm	Legris
Leakage for air	LA	Ø 6 mm	Legris
Leakage for cutting fluid	LKSM	Ø 6 mm	Legris

Dynamic Pressures P2

Oil-air lubrication	4 ±0.2 bar
Air curtain for spindle	2 - 3 bar
Air curtain for coolant through-unit	2 - 3 bar
Tool unclamping	min. 50 bar, max. 60 bar
Taper blowout	5 bar
Coolant motor and bearings	max. 6 bar
Cooling lubricant, external	max. 20 bar
Coolant through-unit, spindle	max. 80 bar
MQL (minimal quantity lubrication)	max. 10 bar
Aerosol	max. 10 bar
Compressed air	max. 10 bar

Storage Conditions

Specified storage temperature	+10° C bis +40° C
Specified storage humidity	max. 55 %rF

Emissions

Permitted vibrations max.	2 mm/s
---------------------------	--------

Miscellaneous

Transport in wooden box, LxBxH	712x342x416 mm
--------------------------------	----------------

9.2 Peripherals Specifications

Compressed Air Supply

Air quality	ISO 8573-1:2010
Class	4.4.4
No. of particles 1 - 5 µm/mm ³	≤ 10000
Pressure dew point	≤ +3°C
Max. oil content	5 mg/m ³
Minimum inlet pressure	0.6 MPa

Air Treatment Unit

Prefilter max. pore size	5 µm
Finefilter pore size	0.5 µm

Hydraulic Power Unit

Operating pressure range	5 - 140 bar
Hydraulic oil	Chapter 9.4

Oil-Air Lubrication Unit

Manufacturer	Fischer								
Oil consumption per bearing	300 mm ³ /h								
Air pressure regulator	min. 5 bar for spindles with DLS lubrication, min. 4 bar for spindles without DLS lubrication, according to flexible tube length & diameter								
Prelubrication time	For details of prelubrication time programming see Para 5.2								
Interval	2 minutes								
Pump Delay Time/Tube Length	<table> <tr> <th>Time</th><th>Length</th></tr> <tr> <td>10 sec</td><td>up to 10 m</td></tr> <tr> <td>15 sec</td><td>10 - 15 m</td></tr> <tr> <td>20 sec</td><td>15 - 40 m*</td></tr> </table>	Time	Length	10 sec	up to 10 m	15 sec	10 - 15 m	20 sec	15 - 40 m*
Time	Length								
10 sec	up to 10 m								
15 sec	10 - 15 m								
20 sec	15 - 40 m*								
Oil pressure switch	external* Art.No 106605								
Air pressure switch	external* Art.No 104431								

* External pressure switches must be fitted on the dosing unit for flexible tube lengths 15 to 40 m.

Oil-Air Mix&Meter Unit (Dosage Unit)

Manufacturerer/Type	Fischer
Number fittings	5
Nozzle	10 mm ³
Oil pressure	min./max. 15 bar/30 bar



Note: For requirements concerning other oil lubrication and dosage units, see operating instructions of the manufacturers concerned.

Cooling motor and bearings with waterbased fluid

Cooling power	6.9 kW
Flowrate min.	8 l/min
Pressure max.	max. 6 bar
Coolant filter fineness	$\beta_{80} \mu\text{m} (c) \geq 75$
Particle cleanliness min.	ISO 4406: 99: Klasse/Class/Classe 23/22/18
Coolant advance temperature	20 - 25° C

Tools & Toolholders

Balance quality rating	G2.5 VDI 2060/ISO 1940
------------------------	------------------------

9.3 Tube Specifications

Coolant IN/OUT, motor and bearings

Material	PEUR	opaque
----------	------	--------

Air curtain for spindle

External tube diameter for

Legris fitting	8.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Oil-Air Lubrication: Mix&Meter Unit to Spindle

External tube diameter for

Legris fitting	4.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Oil-Air Lubrication: Oil Lube Unit to Mix&Meter Unit

External tube diameter for

Legris fitting	6.0 mm +0.05/-0.1
----------------	-------------------

Tube type	flexible highpressure tube
-----------	----------------------------

Working pressure	min. 30 bar
------------------	-------------

Oil-Air Lubrication: Air Treatment Unit to Mix&Meter Unit

External tube diameter for

Legris fitting	8.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Oil return

External tube diameter for

Legris fitting	8.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Tool unclamping

Ermeto fitting	8.0 mm
----------------	--------

Tube type	flexible highpressure tube
-----------	----------------------------

Taper blowout

External tube diameter for

Legris fitting	8.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Coolant through-unit, spindle

Ermeto fitting	8.0 mm
----------------	--------

Tube type	flexible highpressure tube
-----------	----------------------------

Cooling lubricant, external

Tube type	flexible highpressure tube
-----------	----------------------------

Air curtain for coolant through-unit

External tube diameter for	
----------------------------	--

Legris fitting	6.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Leakage for air

External tube diameter for	
----------------------------	--

Legris fitting	6.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

Leakage for cutting fluid

External tube diameter for	
----------------------------	--

Legris fitting	6.0 mm +0.05/-0.1
----------------	-------------------

Material	PUR
----------	-----

9.4 Operating Supplies

Lubricant for Bearings

Quality	DIN 51524, Part 2/ISO VG68 (68mm ² /s by 40°C)
Purity	for prefiltered oil 15/13/10 or higher, acc. to ISO 4406:99
Filtration	β 3μm, min. 100
Recommended	MOTOREX SPINDLE LUBE Hyperclean ISO VG 68
Available at:	www.motorex.com or at Fischer AG (Art. No 879-010-0684)



Unless the lubricants used meet these standards, the lifetime of the precision ball bearings will be shorter or the spindle will fail rapidly.

Hydraulic Oil for Clamping System

Quality	DIN 51524, Part 2/ISO VG32 (32mm ² /s by 40°C)
Purity	18/16/13 or higher, acc. to ISO 4406
Filtration	β 6...10μm, min. 75
Recommended	MOTOREX SPINDLE LUBE Hyperclean ISO VG 32
Available at:	www.motorex.com or at Fischer AG (Art. No 102120).



Unless the lubricants used meet these standards, the lifetime of the precision ball bearings will be shorter or the spindle will fail rapidly.

Lubricant for Clamping Unit

Any hydraulic oil

Coolants for Motor and Bearing

Water-based coolant

Characteristic	free of nitride, corrosion resistant can decompose biologically, maintenance-free
Recommended	MOTOREX COOL-X ready-for-use coolant
Available at:	www.motorex.com or at Fischer AG (Art.No 106717)

Oil-based coolant

Characteristic	corrosion protection low maintenance
Recommended	MOTOREX COOL-OIL ready-for-use coolant
Available at:	www.motorex.com oder bei Fischer AG (Art.No 118179)



Unless coolants are used corresponding to these standards, the life time of some components of the cooling system will be shorter or the spindle will fail. Coolants on glycolbasis can also affect the PEUR tubes and components of zinc.

Cutting fluid for coolant through-unit

Purity	ISO 4406:99 19/17/14
Filtration performance	ISO 16889
Filtration	$\beta (10) \geq 200$

Additif against fungus

Characteristic	preservative agent
Recommendation	Motorex Antisept
available at:	www.motorex.com or at Fischer AG (Art.No 879-900-0100)

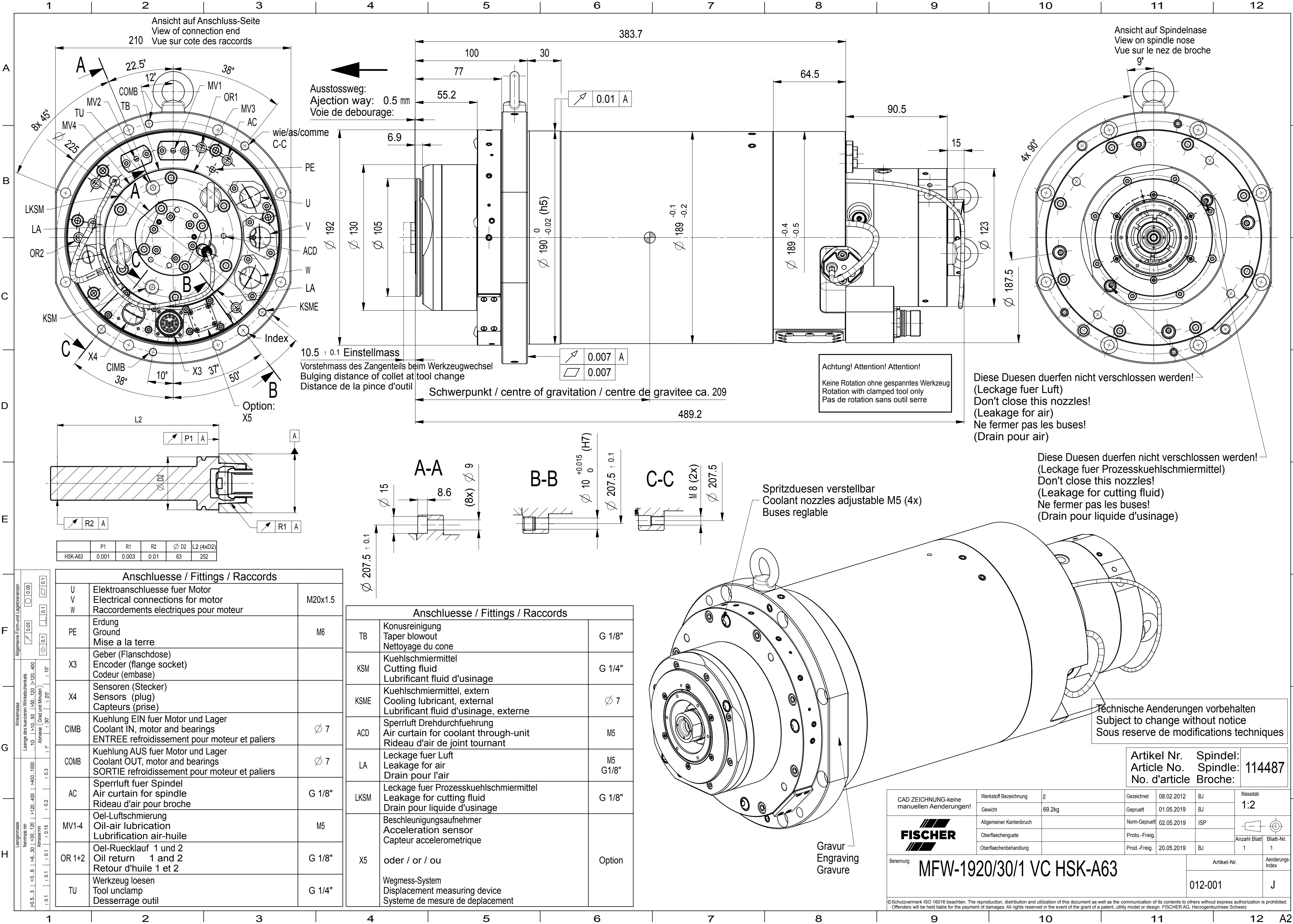
Cleaning additif after fungus attack

Characteristic	cleaning agent
Recommendation	CS-Cleaner
available at:	www.motorex.com or at Fischer AG (Art.No 120891)

10. Annex

10.1 Spindle Drawing

012-001



1
2
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11
12

A
B
C
D
E
F
G
H
A2

Ansicht auf Anschluss-Seite
View of connection end
Vue sur cote des raccords

Ansicht auf Spindelnase
View on spindle nose
Vue sur le nez de broche

Ausstossweg:
Ajection way:
Voie de debourage:

Achtung! Attention! Attention!
Keine Rotation ohne gespanntes Werkzeug
Rotation with clamped tool only
Pas de rotation sans outil serre

Diese Duesen duerfen nicht verschlossen werden!
(Leakage fuer Luft)
Don't close this nozzles!
(Leakage for air)
Ne fermer pas les buses!
(Drain pour air)

Diese Duesen duerfen nicht verschlossen werden!
(Leakage fuer Prozesskuehlschmiermittel)
Don't close this nozzles!
(Leakage for cutting fluid)
Ne fermer pas les buses!
(Drain pour liquide d'usinage)

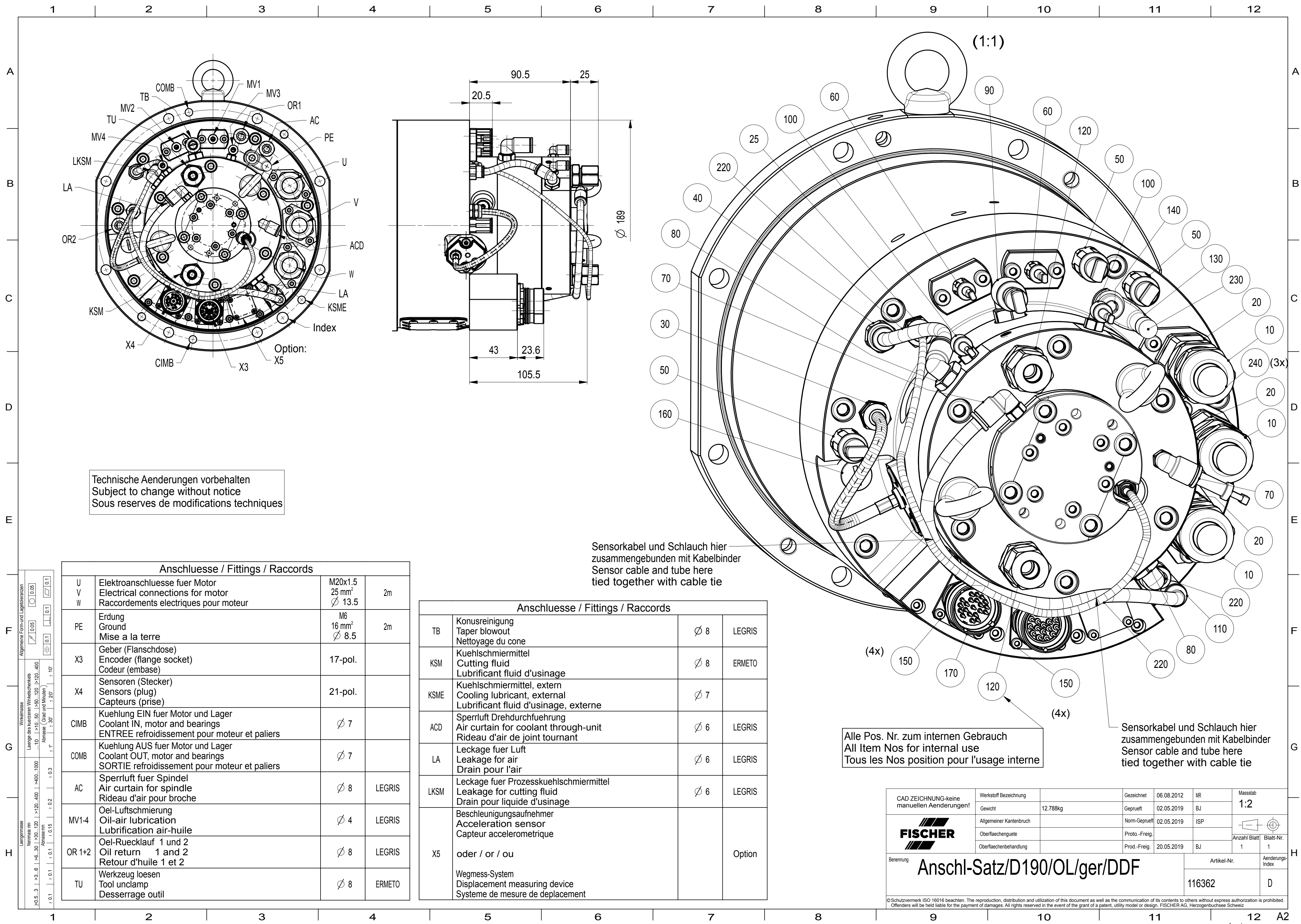
Technische Aenderungen vorbehalten
Subject to change without notice
Sous reserve de modifications techniques

Artikel Nr. Spindel:
Article No. Spindle: 114487
No. d'article Broche:

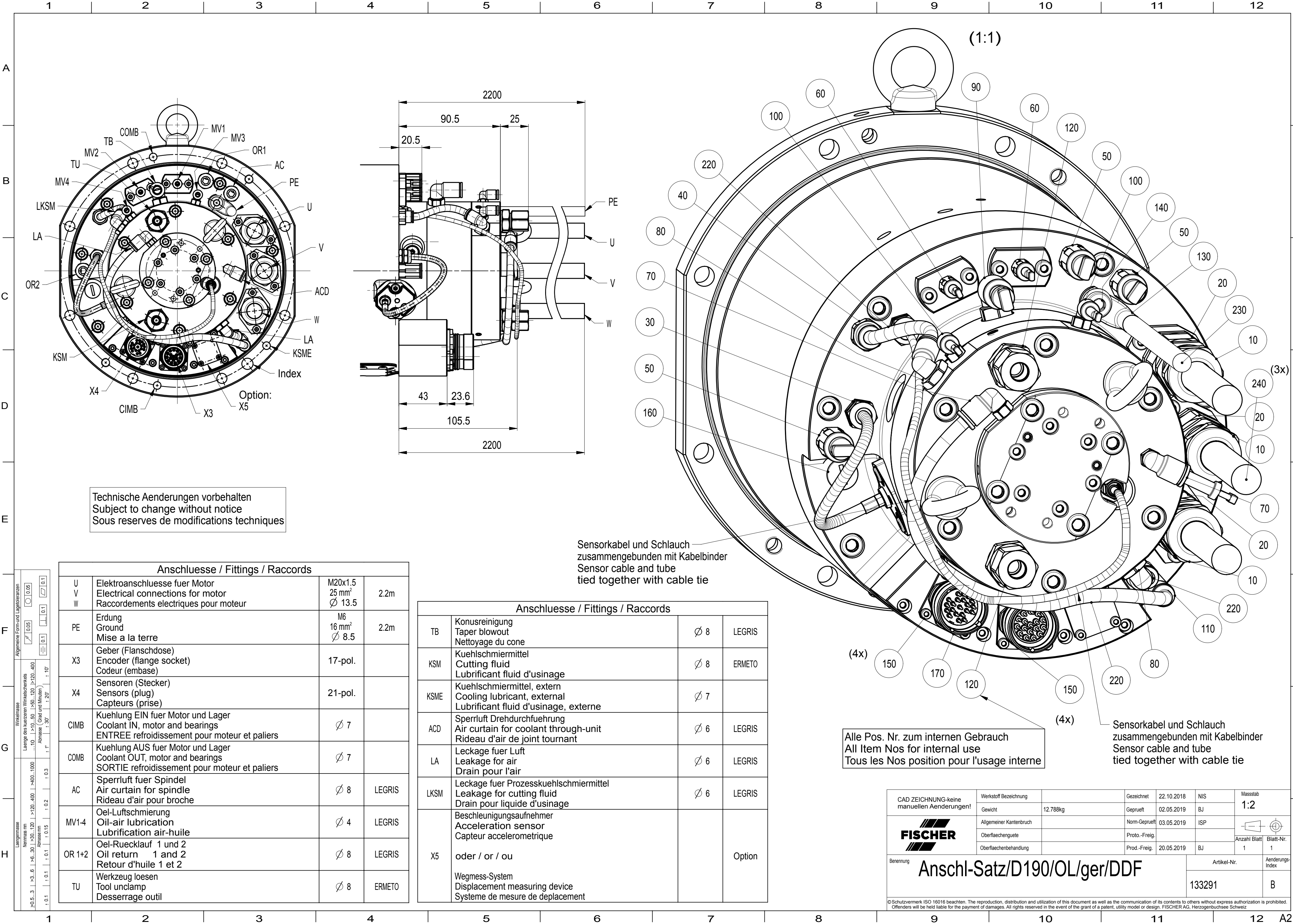
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	Gewicht	69.2kg	Geprueft	01.05.2019	BJ		
	Allgemeiner Kantenbruch		Norm-Geprueft	02.05.2019	ISP		
	Oberflaechengüte		Proto.-Freig.			Anzahl Blatt	Blatt-Nr.
	Oberflaechenbehandlung		Prod.-Freig.	20.05.2019	BJ	1	1
Benennung	MFW-1920/30/1 VC HSK-A63					Artikel-Nr.	Aenderungs- Index
						012-001	J

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10.2 Fitting Set 116362



10.3 Fitting Set 133291



Technische Aenderungen vorbehalten
Subject to change without notice
Sous reserves de modifications techniques

Sensorkabel und Schlauch
zusammengebunden mit Kabelbinder
Sensor cable and tube
tied together with cable tie

Alle Pos. Nr. zum internen Gebrauch
All Item Nos for internal use
Tous les Nos position pour l'usage interne

Sensorkabel und Schlauch
zusammengebunden mit Kabelbinder
Sensor cable and tube
tied together with cable tie

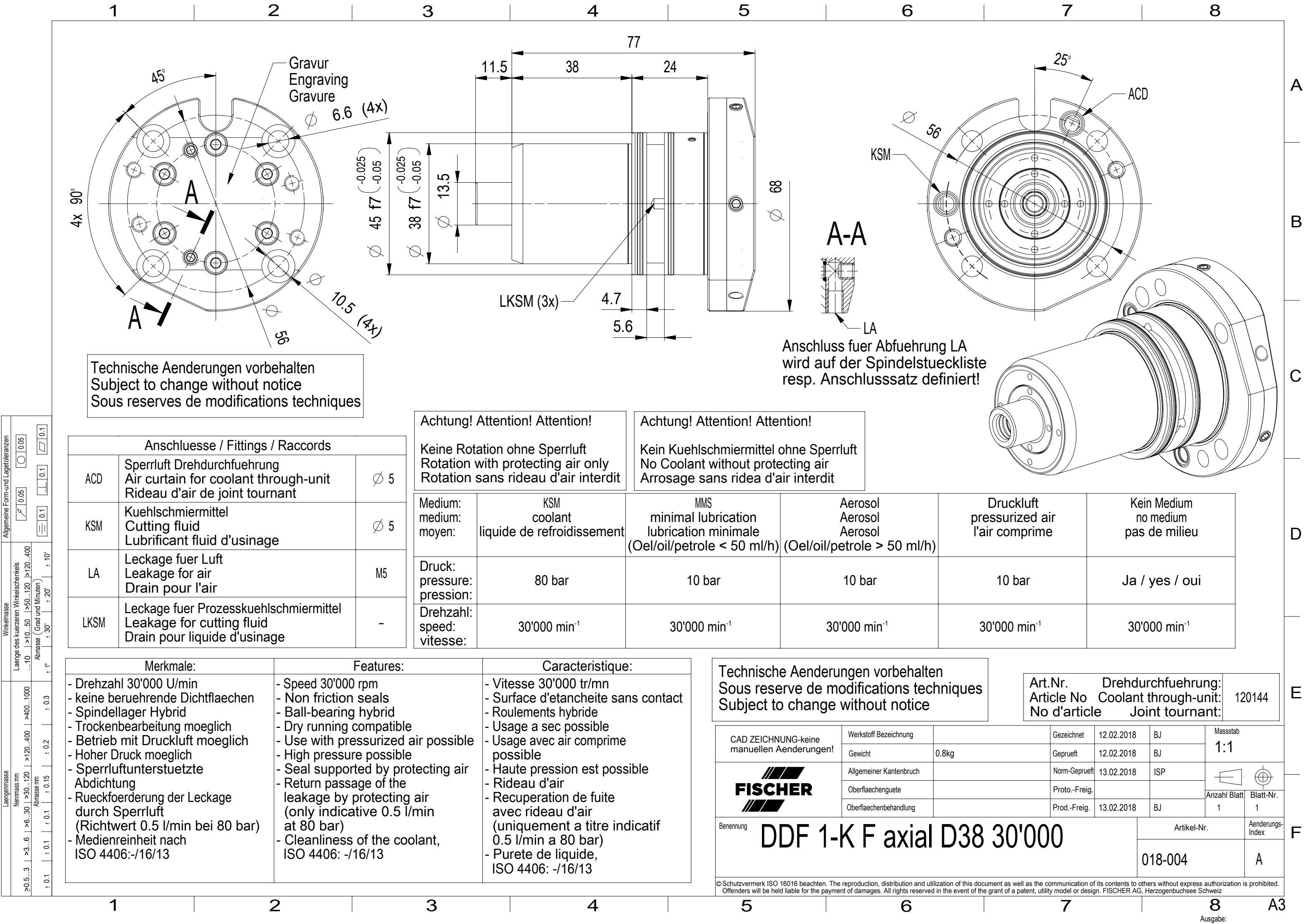
Anschuesse / Fittings / Raccords			
U V W	Elektroanschluesse fuer Motor Electrical connections for motor Raccordements electriques pour moteur	M20x1.5 25 mm ² Ø 13.5	2.2m
PE	Erdung Ground Mise a la terre	M6 16 mm ² Ø 8.5	2.2m
X3	Geber (Flanschdose) Encoder (flange socket) Codeur (embase)	17-pol.	
X4	Sensoren (Stecker) Sensors (plug) Capteurs (prise)	21-pol.	
CIMB	Kuehlung EIN fuer Motor und Lager Coolant IN, motor and bearings ENTREE refroidissement pour moteur et paliers	Ø 7	
COMB	Kuehlung AUS fuer Motor und Lager Coolant OUT, motor and bearings SORTIE refroidissement pour moteur et paliers	Ø 7	
AC	Sperrluft fuer Spindel Air curtain for spindle Rideau d'air pour broche	Ø 8	LEGRIS
MV1-4	Oel-Luftschmierung Oil-air lubrication Lubrification air-huile	Ø 4	LEGRIS
OR 1+2	Oel-Ruecklauf 1 und 2 Oil return 1 and 2 Retour d'huile 1 et 2	Ø 8	LEGRIS
TU	Werkzeug loesen Tool unclamp Desserrage outil	Ø 8	ERMETO

Anschuesse / Fittings / Raccords			
TB	Konusreinigung Taper blowout Nettoyage du cone	Ø 8	LEGRIS
KSM	Kuehlschmiermittel Cutting fluid Lubrifiant fluid d'usinage	Ø 8	ERMETO
KSME	Kuehlschmiermittel, extern Cooling lubricant, external Lubrifiant fluid d'usinage, externe	Ø 7	
ACD	Sperrluft Drehdurchfuehrung Air curtain for coolant through-unit Rideau d'air de joint tournant	Ø 6	LEGRIS
LA	Leckage fuer Luft Leakage for air Drain pour l'air	Ø 6	LEGRIS
LKSM	Leckage fuer Prozesskuehlschmiermittel Leakage for cutting fluid Drain pour liquide d'usinage	Ø 6	LEGRIS
X5	Beschleunigungsaufnehmer Acceleration sensor Capteur accelerometrique		
	oder / or / ou		Option
	Wegmess-System Displacement measuring device Systeme de mesure de deplacement		

	CAD ZEICHNUNG-keine manuellen Aenderungen!	Werkstoff Bezeichnung	Gezeichnet	22.10.2018	NIS	Massstab 1:2
		Gewicht	Geprueft	02.05.2019	BJ	
		Allgemeiner Kantenbruch	Norm-Geprueft	03.05.2019	ISP	
		Oberflaechenguete	Proto.-Freig.			
		Oberflaechenbehandlung	Prod.-Freig.	20.05.2019	BJ	Anzahl Blatt 1
						Blatt-Nr. 1
Benennung Anschl-Satz/D190/OL/ger/DDF					Artikel-Nr. 133291	
					Aenderungs- Index B	

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10.4 Coolant through-unit 018-004



10.5 Peripherals Connection Diagram

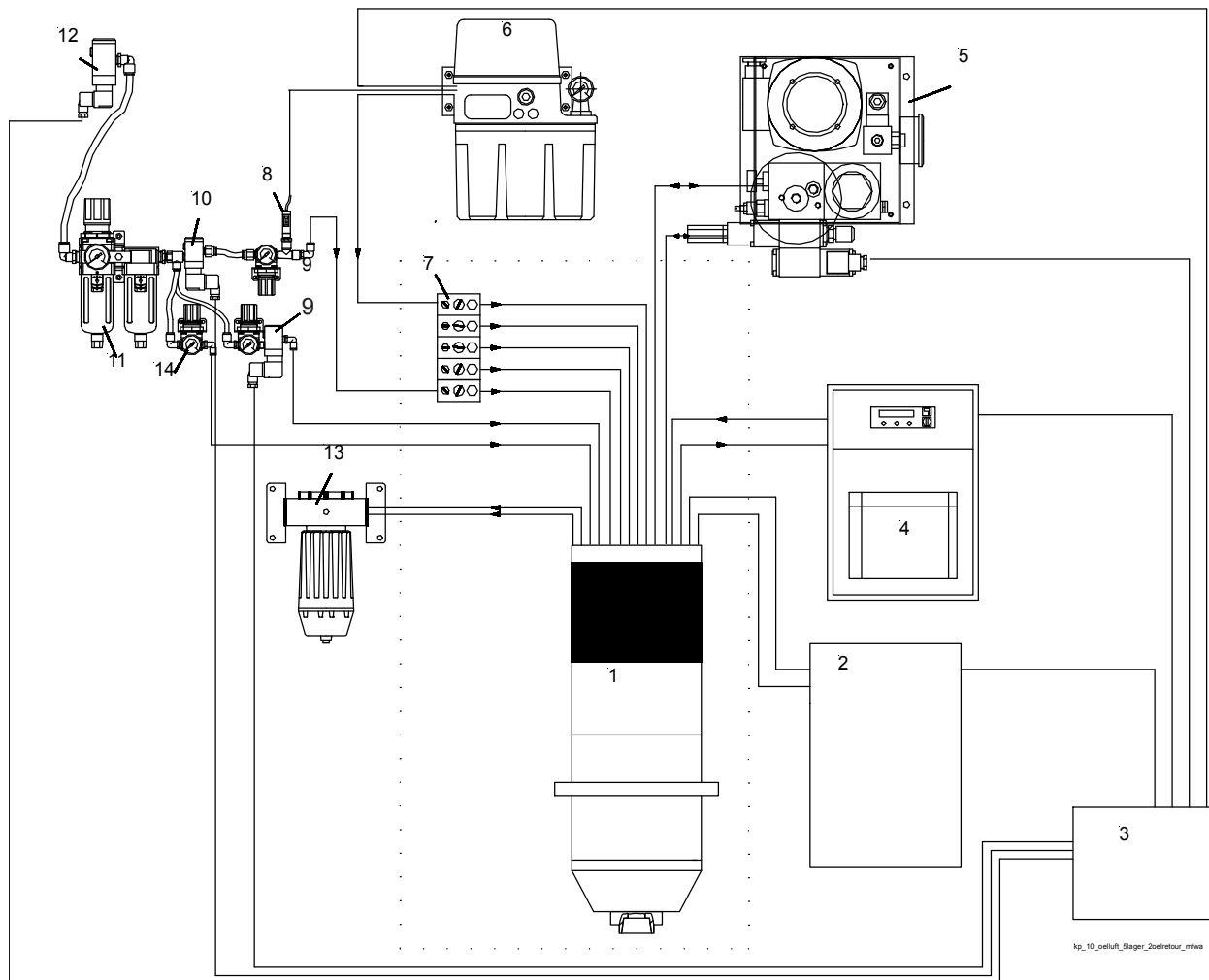


Fig. 10 - 1: Peripherals Connection Diagram

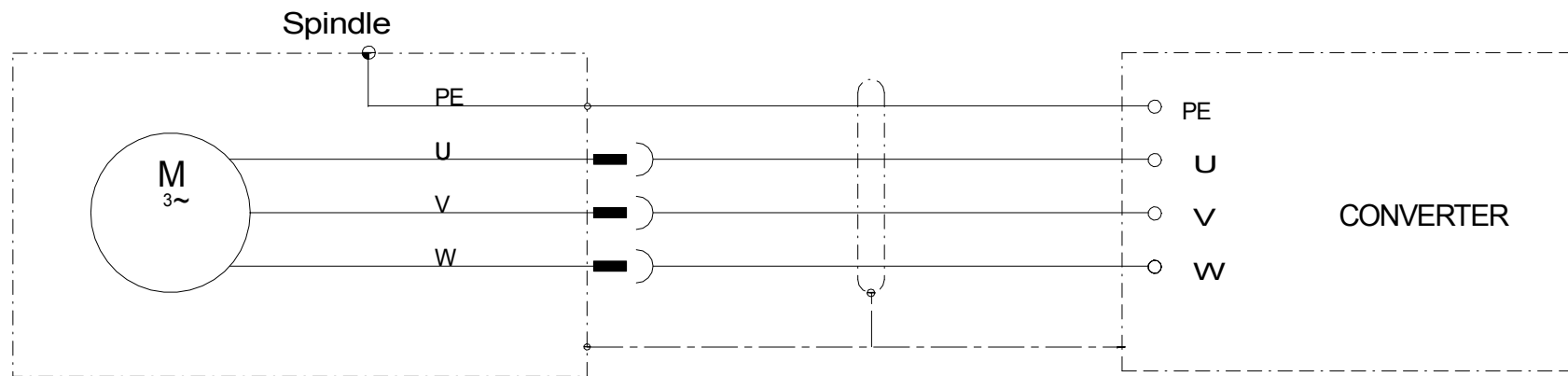
Legend

- | | |
|-----------------------------------|--|
| 1 Spindle | 8 Pressure switch for oil-air lubrication |
| 2 Frequency converter | 9 3/2 way valve for taper blow out |
| 3 Stored programmed control (SPC) | 10 3/2 way valve for air in oil-air lubrication system |
| 4 Cooling system | 11 Air treatment unit |
| 5 Hydraulic power unit | 12 3/2 way valve (main valve) |
| 6 Oil-air lubrication unit | 13 Oil separator |
| 7 Dosage unit | 14 Pressure control for curtain air |

10.6 Electrical Schematics

10.6.1 Motor

102469

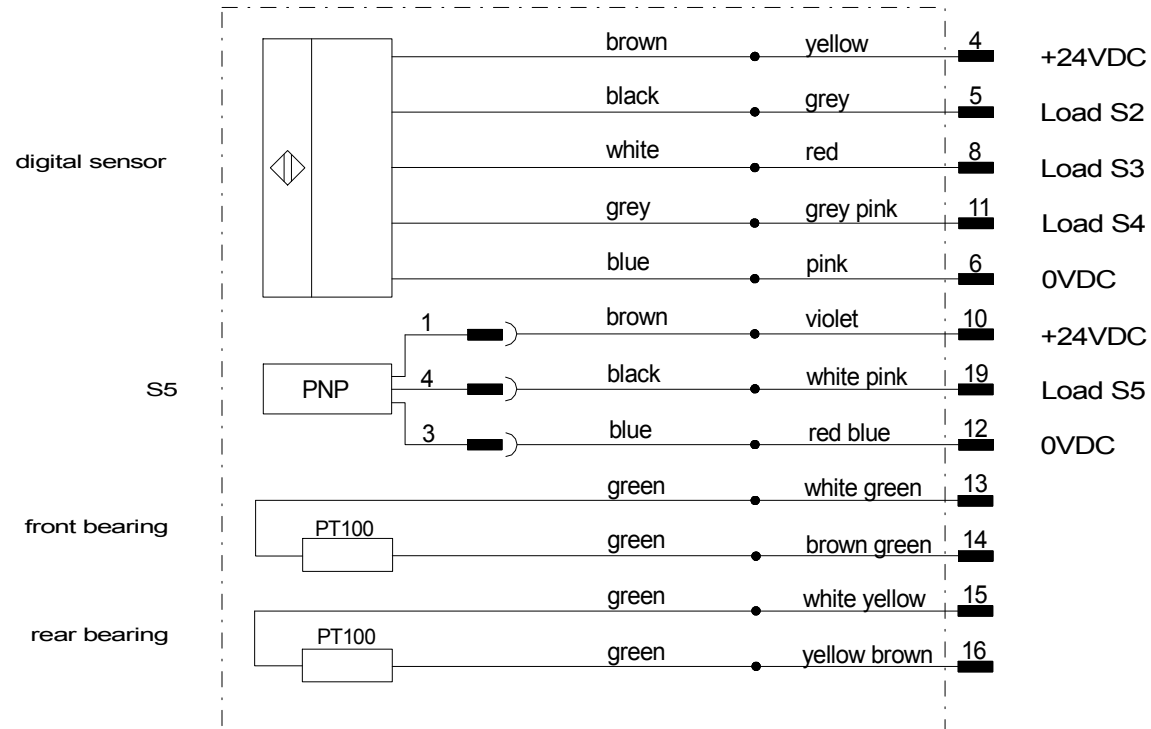


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Checked	06.01.2003	Pu						
Released	06.01.2003	glo		© All rights reserved. Duplication or transmission in any form without written permission of the Fischer AG Herzogenbuchsee Switzerland are prohibited.	Remarks	Drawing number 102469	Index A	Total 1
Draft								Page 1

10.6.2 Proximity Switches (Sensors) and temperature control

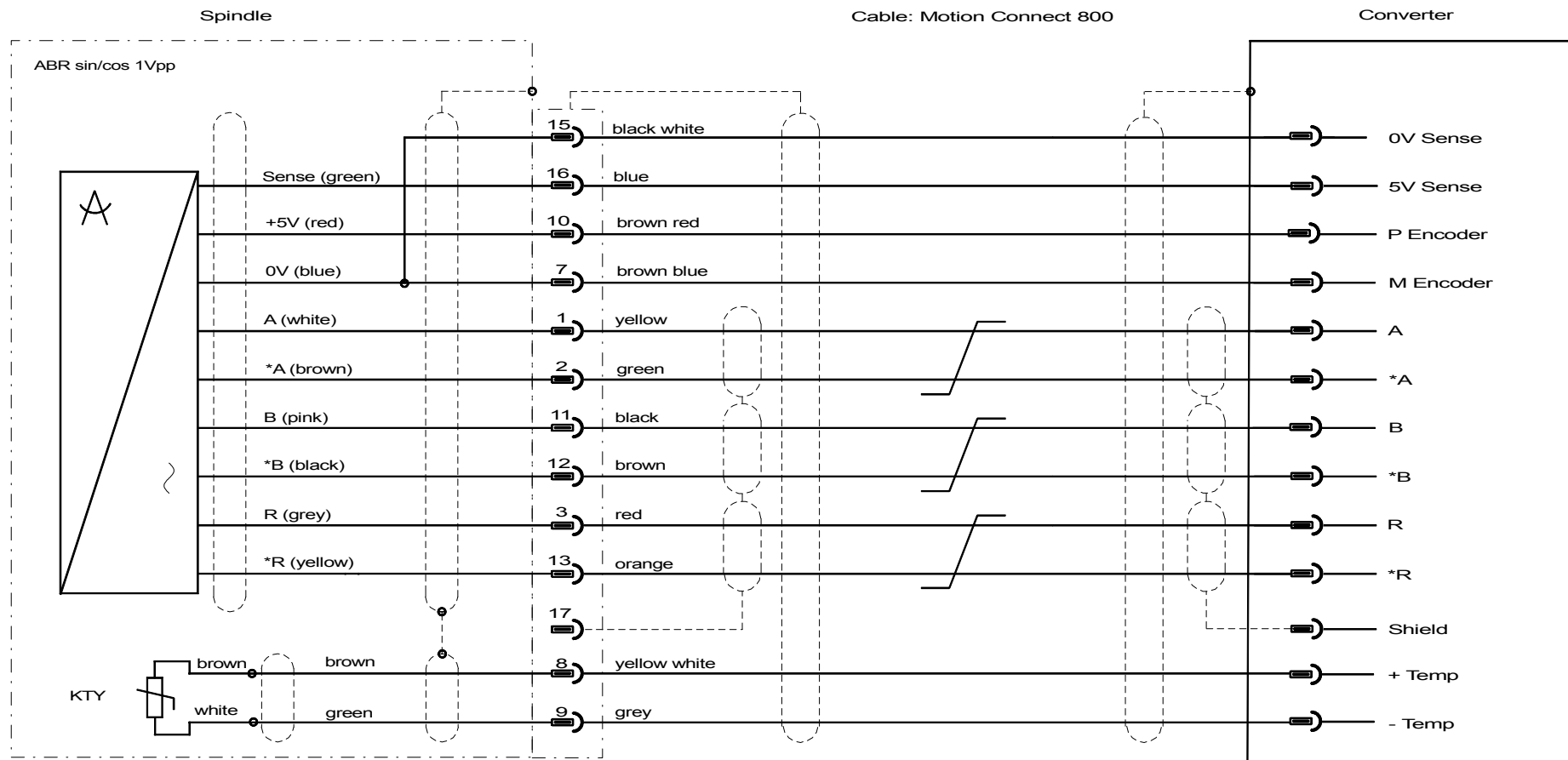
92000044

Spindle



10.6.3 Encoder + KTY

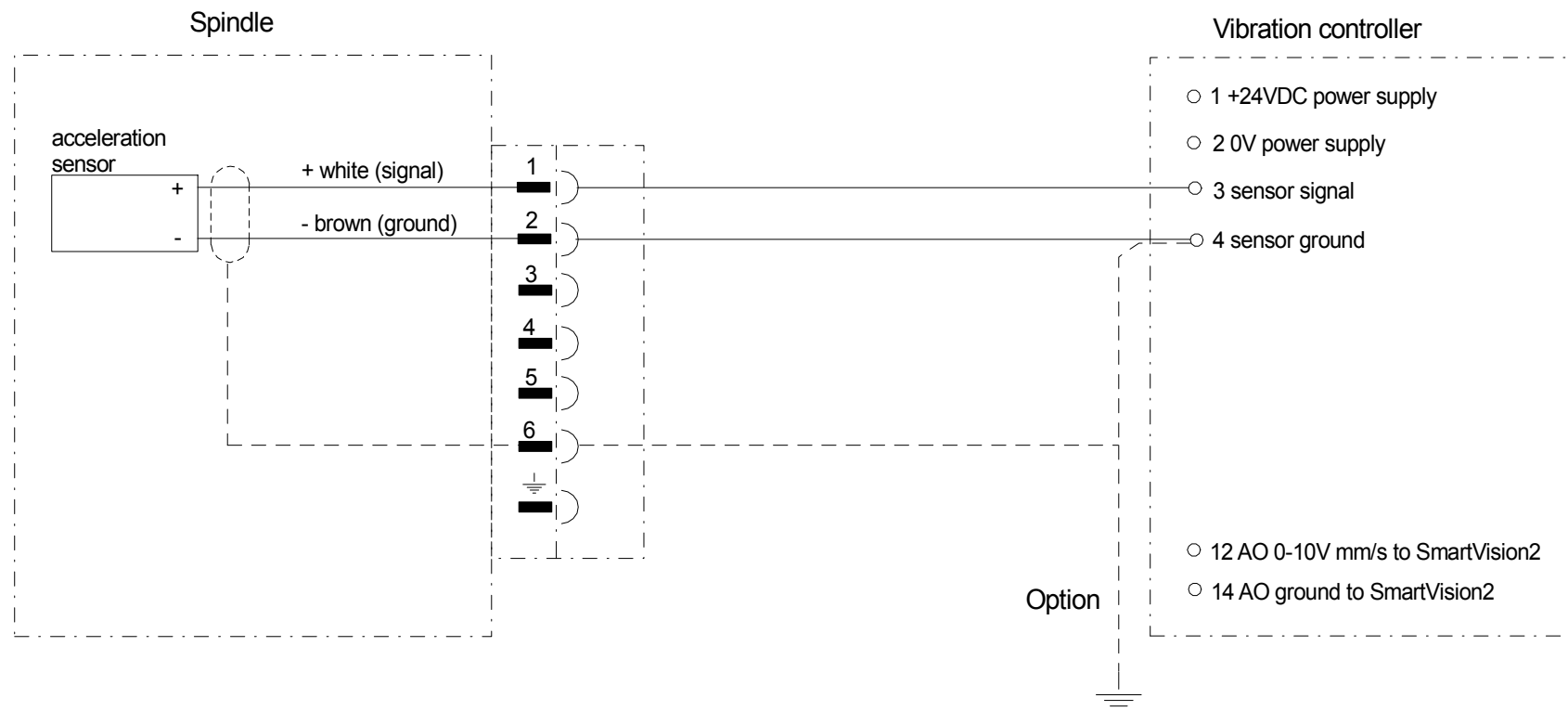
91170-183



Designed	25.02.2013	FA		Project description		Page description					
Checked	25.02.2013	BE		ES GK LB VCS 17P KTY							
Released	25.02.2013	FA				Remarks		Drawing number	Index	Total	1
Draft				© All rights reserved. Duplication or transmission in any form without written permission of the Fischer AG Herzogenbuchsee Switzerland are prohibited.				91170-183	L	Page	1

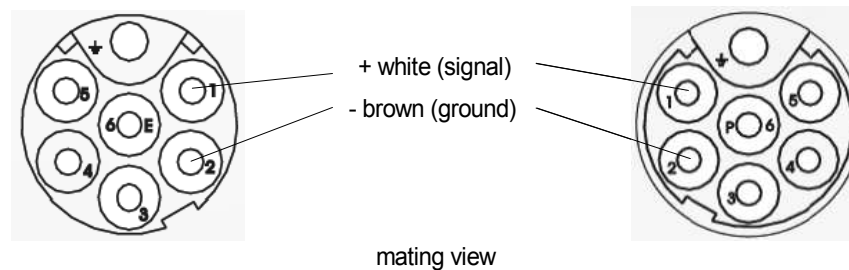
10.6.4 Option Acceleration Sensor 121041

92000007



Plug Speedtec 863 Contact Pin

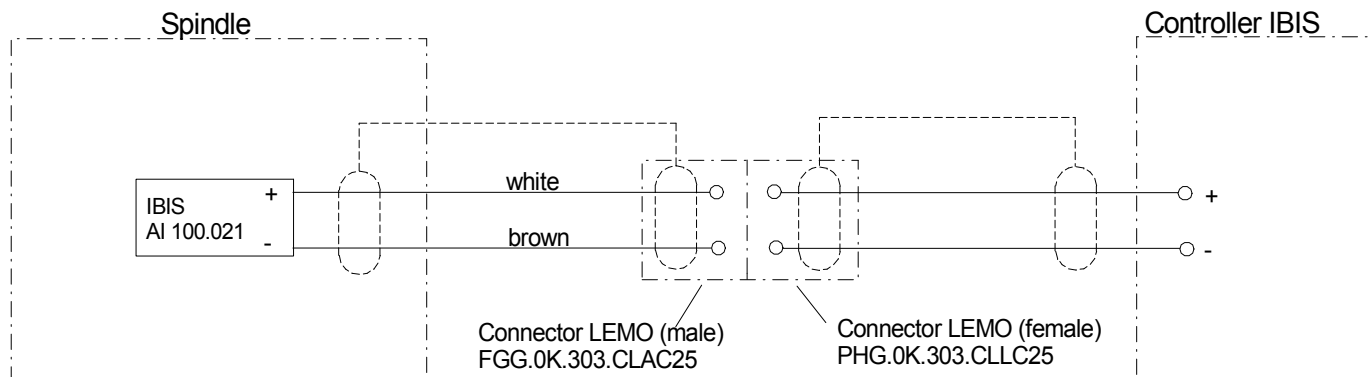
Receptacle Straight Speedtec 861 Contact Socket



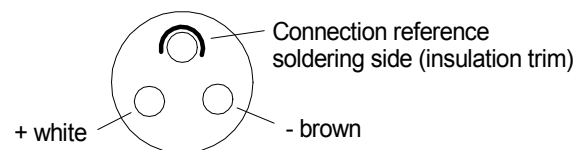
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Released	20.02.2012	FA		© All rights reserved. Duplication or transmission in any form without written permission of the Fischer AG Herzogenbuchsee Switzerland are prohibited.		Remarks	Drawing number	Index	Total
Draft							92000007	C	1
								Page	1

10.6.5 Option Acceleration Sensor 113262

106386 / 104386



Connector LEMO (male)
FGG.0K.303.CLAC25



Designed	12.03.2019	GRC		Project description	Page description			
Checked	12.03.2019	HM		ES S IBIS BeschI-Aufn				
Released	12.03.2019	GRC			Remarks	Drawing number	Index C	Total 1
Draft				© All rights reserved. Duplication or transmission in any form without written permission of the Fischer AG Herzogenbuchsee Switzerland are prohibited.		106386		Page 1

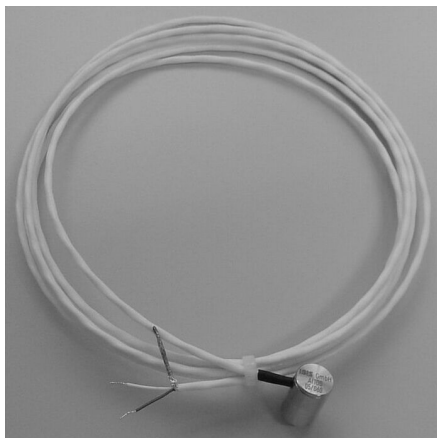
Vibrationsüberwachung

Vibration monitoring

Contrôle de vibration

(Sensor/ Auswertelektronik/ Hersteller)
(sensors/ evaluation electronics/ manufacturer)
(sensors/ électronique d'évaluation/ fabricant)
-AI100/ SiViB1000/ IBIS

Photo



Pin Assignment

Evaluation electronics

Binder	Function
1	+ 24 V DC
2	Ground 24 VDC
3	Sensor signal
4	Sensor ground
5	Sensor error Out (5 V TTL)
6	Sensor shield
7	Alarm A
8	Alarm B
9	Alarm C
10	Threshold value input
11	Ground alarms
12	Alarms
13	Analogue output 4-20 mA mm/s (Option)
14	Analogue output out 4-20 mA gSE (Option)
15	Reset

Sensors

Connector	Function
White	Signal
Brown	Ground
Shield	Shield

Sensorkabel:	Länge darf verändert werden (Spannungsverluste sind zu berücksichtigen) Eine zusätzliche Steckverbindung ist zulässig
Anschlusskabel:	Längen dürfen verändert werden (Spannungsverluste sind zu berücksichtigen)
Querschnitte:	Nicht kleiner als die vom Hersteller verwendeten
Schirm:	An geeigneter Stelle mit Erde verbinden (verbesserte Störungsunterdrückung) Bei Verwendung von zusätzlichen Steckverbindungen in der Sensorleitung darf der Schirm nicht unterbrochen werden
Pinbelegung:	-
Auswerteinheit:	Wird an CNC angeschlossen Bei Verwendung von Smart Vision erfolgt Anschluss an CNC via Smart Vision

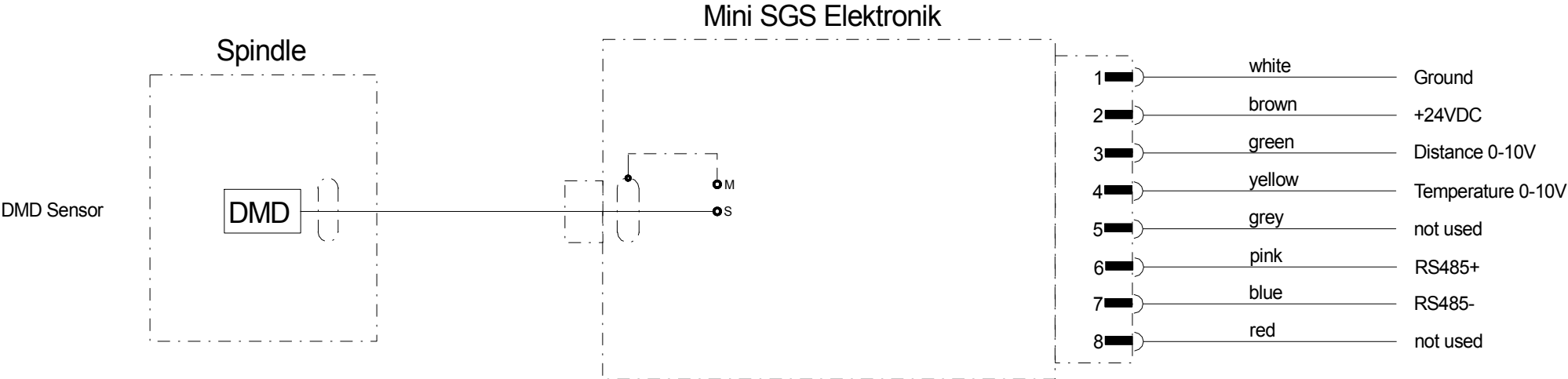
Sensor cable:	Length may be changed (voltage losses have to be considered) One additional plug is acceptable
Sensor cable:	Length may be changed (voltage losses have to be considered)
Cross sectional area:	Not smaller than the ones used by the manufacturer
Shield:	Connect to earth in an appropriate place (improved noise suppression) If using additional connectors in the sensor cable shield must not be interrupted
Pin assignment:	-
Electronic unit:	Has to be connected to the CNC When using Smart Vision connection (of electronic unit) to the CNC occurs via Smart Vision

Câble de sonde:	La longueur peut être modifiée (des pertes de tension doivent être prises en considération) un connecteur supplémentaire, est admissible
Câble de raccordement :	La longueur peut être modifiée (des pertes de tension doivent être prises en considération)
Section câble :	Pas plus petit que ceux du fabricant
Blindage:	Connecteur à la place appropriée à la terre (réduction des parasites) En cas d'utilisation de connecteurs supplémentaires sur le câble du détecteur, le blindage ne doit pas être interrompu.
Brochage:	-
Unité de mesure:	à raccorder à la CNC Lors de l'utilisation de SmartVision, il faut faire le raccordement à la CNC via SmartVision

Designed	12.01.2005	pz		Project description		Page description			
Checked	28.01.2005	pz		AS-Vib.sys IBIS AI100					
Released	28.01.2005	pz				Remarks		Drawing number	
Draft				© All rights reserved. Duplication or transmission in any form without written permission of the Fischer AG Herzogenbuchsee Switzerland are prohibited.				104386	
								Index	A
								Total	1
								Page	1

10.6.6 Option DMD/SGS Mini

92000066



Designed	12.03.2013	GRC
Checked	12.03.2013	FA
Released	12.03.2013	GRC
Draft		



Project description
ES S 2P DMD
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Page description
Remarks
Drawing number
92000066

Index	B	Total	1
		Page	1

10.4.7 Drive Parameters (SIN) and OFF-diagram

106944

Number / Name: 106944 A SIE D 20/30 011		Antriebsdaten Asynchronmotor drive parameters asynchronous motor paramètres moteur asynchrone		
Index: D				
Issued: 12.01.2012 / HM				
Checked: 12.01.2012 / HM				
Approved: 12.01.2012 / HM				
Changed: 04.12.2018 / HM				
Parameterliste Sinamics / parameter list Sinamics / liste de paramètres Sinamics				
Motor- und Spindeldata / motor and spindle data / données du moteur et de la broche				
Artikel-Nr. Motor / article-Nr. motor / art. N° moteur		106944		
Motordrossel / motor choke / self moteur		0.05 mH		
Zwischenkreisspannung / interm. circuit voltage / tension circuit int.		600 V		
Polzahl Motor / number of poles / nombre de pôles		4		
Schaltungsart / connection method / mode de connexion		delta connection		
Datenursprung / data source / origin des données		optimization at load condition		
Empfohlene Hard- und Software / recommended hard- and software / hard- et software recommandé				
Kontrolleinheiten / control units / unités de contrôle		CU320, NX10, NX15 - check with Siemens software "Sizer"!		
Leistungsteil / power module / module de puissance		200/230/282 6SL312X-1TE32-0AXX		
840D sl CNC SW-version		4.7 HF17 or higher		
Type / PAR	Index			
Assistant				
D p1300	[all]	21 ---	Regelungsart / Control method / Type de régulation	
---	---	DC 510 - 720 V	Anschlussspannung / Connection voltage / Tension d'alimentation	
---	---	ja / yes / oui	Motordaten eingeben / Enter motor data / Saisir les paramètres moteur	
M p300	[all]	1 ---	Motortyp / Motor type / Type de moteur	
---	---	ja / yes / oui	Wollen Sie mechanische Daten eingeben / Do you want to enter the mechanical data / Voulez-vous entrer des données mécaniques	
---	---	ja / yes / oui	Wollen Sie die Ersatzschaltbilddaten eingeben / Do you want to enter the equivalent circuit diagram data ? / Voulez-vous saisir les paramètres du schéma équivalent ?	
D p340	[all]	vollst ohne ESB / comp. without / Cal. com. sens	Berechnung der Motor-/Reglerdaten / Calculation of the motor/controller data / Calcul des paramètres moteur / régulateur	
---	---	Daten eingeben / Enter data / Saisir param.	Geberdaten / Encoder data / Param. Codeur	
p500	---	Spindelantrieb / Spindle drive / Entr. de broche	Technologische Applikation / Technological application / Application technologique	
p922	---	Freie Teleg. / Free telegr. / Conf. de T. libre	PROFIdrive Telegramm / PROFIdrive message frame / Télégramme PROFIdrive	
Main				
p115	[0]	62.50 µs	Stromreglertakt / Sampling time current controller / Cycle rég. courant	
p115	[1]	500.00 µs	Drehzahlreglertakt / Sampling time speed controller / Cycle rég. Vitesse	
p115	[2]	500.00 µs	Flussreglertakt / Sampling time flux controller / Cycle rég. flux	
p130	---	1 ---	Motordatensatz (MDS) Anzahl / Number of Motor Data Sets (MDS) / Jeux de paramètres moteur (MDS) Nombre	
p2000	---	30'000 1/min	Bezugsdrehzahl Bezugsfrequenz / Reference speed reference frequency / Vitesse de rotation de référence Fréquence de référence	
p3510	[0]	600 V	Einspeisung Zwischenkreisspannung Sollwert / Infeed DC link voltage setpoint / Alimentation Tension de circuit intermédiaire Consigne	
DDS				
D p186	[0] ---	0 ---	Motordatensatz (MDS) Nummer / Motor Data Sets (MDS) number / Jeu de paramètres moteur (MDS) Numéro	
D p640	[0] ---	200 --- Aeff	Stromgrenze / Current limit / Limite de courant	
D p1082	[0] ---	30'000 --- 1/min	Maximaldrehzahl / Maximum speed / Vitesse maximale	
D p1441	[0] ---	1.0 --- ms	Drehzahlwert Glättungszeit / Actual speed smoothing time / Mesure de vitesse Temps de lissage	
D p1520	[0] ---	38.38 --- Nm	Drehmomentgrenze oben/motorisch / Torque limit upper/motoring / Limite de couple supérieure/en moteur	
D p1521	[0] ---	-38.38 --- Nm	Drehmomentgrenze unten/generatorisch / Torque limit lower/regenerative / Limite de couple inférieure/en génératrice	
D p1530	[0] ---	78.75 --- kW	Leistungsgrenze motorisch / Power limit motoring / Limite de puissance en fonctionnement en moteur	

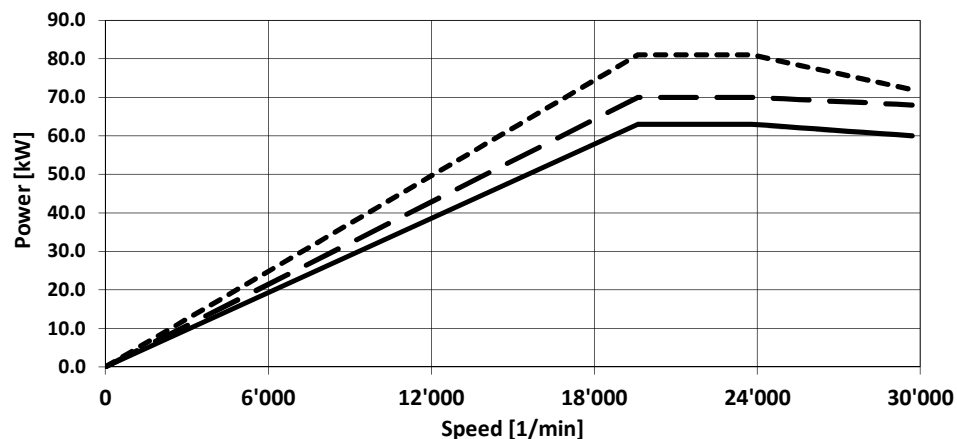
Number / Name:		106944 A SIE D 20/30 011	Antriebsdaten Asynchronmotor drive parameters asynchronous motor paramètres moteur asynchrone	
Index:		D		
Issued:		12.01.2012 / HM		
Checked:		12.01.2012 / HM		
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Changed:		04.12.2018 / HM		
Parameterliste Sinamics / parameter list Sinamics / liste de paramètres Sinamics				
D p1531	[0] ---	-78.75 kW --- kW	Leistungsgrenze generatorisch / Power limit regenerating / Limite de puissance en fonctionnement en génératrice	
D p1800	[0] ---	8.000 --- kHz	Frequenz Pulsbreitenmodulation / Pulse width modulation frequency / Fréq. modulation larg. Impulsion	
D p2155	[0] ---	210'000 --- 1/min	Drehzahlschwellwert 2 / Speed threshold 2 / Seuil de vitesse 2	
CDS				
C p820	[0]	0 ---	BI: Antriebsdatensatz-Anwahl DDS Bit 0 / BI: Drive data set selection DDS bit 0 / BI: Sélection du jeu de paramètres d'entraînement DDS Bit 0	
C p821	[0]	0 ---	BI: Antriebsdatensatz-Anwahl DDS Bit 1 / BI: Drive data set selection DDS bit 1 / BI: Sélection du jeu de paramètres d'entraînement DDS Bit 1	
C p822	[0]	0 ---	BI: Antriebsdatensatz-Anwahl DDS Bit 2 / BI: Drive data set selection DDS bit 2 / BI: Sélection du jeu de paramètres d'entraînement DDS Bit 2	
C p823	[0]	0 ---	BI: Antriebsdatensatz-Anwahl DDS Bit 3 / BI: Drive data set selection DDS bit 3 / BI: Sélection du jeu de paramètres d'entraînement DDS Bit 3. ATTENTION: Bit must be modified from PLC during handling of the Y-D connection change. For Y-connection, bit must be set to value of 1. D-connection requires value of 0.	
C p824	[0]	0 ---	BI: Antriebsdatensatz-Anwahl DDS Bit 4 / BI: Drive data set selection DDS bit 4 / BI: Sélection du jeu de paramètres d'entraînement DDS Bit 4	
EDS				
---	---	Ink. Sin;Cos / Inc sin;cos /	Messsystem / Encoder Type / Type de codeur	
E p404	[0].12	Nullm.:Umd. / 1 zero	Nullmarken Konfiguration / Zero marks configuration / Tops zéro config.	
E p408	[0]	256 ---	Geberstrichzahl Motormesssystem / No.encoder marks, motor meas.System / Nb traits capt. syst. mesure mot.	
MDS				
---	[0] ---	delta ---	Schaltungsart / connection method / mode de connexion	
M p304	[0] ---	380 --- V _{eff}	Motor-Bemessungsspannung / Rated motor voltage / Tension assignée du moteur	
M p305	[0] ---	133.00 --- A _{eff}	Motor-Bemessungsstrom / Rated motor current / Courant assigné du moteur	
M p307	[0] ---	63.00 --- kW	Motor-Bemessungsleistung / Rated motor power / Puissance assignée du moteur	
M p308	[0] ---	0.790 --- cosφ	Motor-Bemessungsleistungsfaktor / Rated motor power factor / Facteur de puissance assignée du moteur	
M p310	[0] ---	660.00 --- Hz	Motor-Bemessungsfrequenz / Rated motor frequency / Fréquence assignée du moteur	
M p311	[0] ---	19'600 --- 1/min	Motor-Bemessungsdrehzahl / Rated motor speed / Vitesse assignée du moteur	
M p320	[0] ---	61.00 --- A _{eff}	Motor-Bemessungsmagnetisierungsstrom / Motor rated magnetization current/short-circuit current / Courant assigné de magnétisation/de court-circuit du moteur	
M p322	[0] ---	30'000 --- 1/min	Maximale Motordrehzahl / Maximum motor speed / Moteur vitesse maximale	
M p326	[0] ---	100 --- %	Motor-Kippmomentkorrekturfaktor / Motor stall torque correction factor / Facteur de correction de couple de décrochage du moteur	
M p335	[0] ---	Water cooling ---	Motorkühlart / Motor cooling type / Mode de refroidissement du moteur	
M p341	[0] ---	0.014590 --- kgm²	Motorträgheitsmoment / Moment of inertia of motor / Moment inertie moteur	
M p342	[0] ---	1.000 ---	Trägheitsmoment Verhältnis Gesamt zu Motor / Ratio between the total and motor moment of inertia / Moment inertie Rapport total / moteur	
M p344	[0] ---	16.0 --- kg	Motorgewicht / Motor weight / Poids du moteur	
M p348	[0] ---	19'600 --- 1/min	Einsatzdrehzahl Feldschwächung / Speed at the start of field weakening / Vitesse de transition Réduction du champ	
M p350	[0] ---	0.014 --- Ω	Motor-Ständerwiderstand kalt / Motor stator resistance, cold / Moteur Résistance stator à froid	
M p352	[0] ---	0.000 --- Ω	Leitungswiderstand / Cable resistance / Résistance du câble	

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Parameterliste Sinamics / parameter list Sinamics / liste de paramètres Sinamics				
M p353	[0] ---	0.050 --- mH	Induktivität der Vorschaltddrossel / Series reactor inductance / Inductance réactance en amont	
M p354	[0] ---	0.017 --- Ω	Motor-Läuferwiderstand kalt / Motor rotor resistance cold / Résistance de rotor du moteur à froid	
M p356	[0] ---	0.039 --- mH	Motor-Ständerstreuinduktivität / Motor stator leakage inductance / Moteur Inductance de fuite du stator	
M p358	[0] ---	0.031 --- mH	Läuferstreureaktanz / Rotor leakage reactance / Réactance fuite rotor / Moteur Inductance stator Axe	
M p360	[0] ---	0.854 --- mH	Hauptfeldreaktanz / Magnetizing reactance / Inductance principale moteur	
M p600	[0] ---	1 ---	Motortemperatursensor für Überwachung / Motor temperature sensor for monitoring / Sonde thermométrique du moteur pour surveillance 1 = connected to smc20, 11 = connect to drive	
M p601	[0] ---	check fn ---	Temperatursensor Sensortyp / Temperature sensor, sensor type / Sonde thermoélectrique du moteur Type de sonde 1 = PTC, 2 = KTY, 6 = PT1000, check fn --> check footnote 1)	
M p603	[0] ---	0 ---	CI: Motortemperatur Signalquelle / CI: Motor temperature signal source / CI: Température moteur Source de signal	
M p604	[0] ---	150 --- °C	Motorübertemperatur Warnschwelle / Motor overtemperature alarm threshold / Echauffement du moteur Seuil d'alarme	
M p605	[0] ---	155 --- °C	Motorübertemperatur Störschwelle / Motor overtemperature fault threshold / Echauffement du moteur Seuil de défaut	
M p606	[0] ---	240 --- s	Motorübertemperatur Zeitstufe / Motor overtemperature timer / Echauffement du moteur temorisation	
M p620	[0] ---	2 ---	Thermische Adaption Ständer- und Läuferwiderstand / Thermal adaptation, stator and rotor resistance / Adaptation thermique Résistance rotor et stator	
Reglerparameter mit Drossel 0.05mH / Controller parameters with choke 0.05mH / Paramètres du régulateur avec self 0.05mH				
p118	---	0.00 μs	Stromregler Rechentotzeit / Current controller computing dead time / Régulateur de courant Temps mort calcul	
D p1460	[0] ---	2.700 Nms/Rad	Drehzahlregler P-Verstärkung Adaptiondrehzahl unten / Speed setpoint filter 1 time constant / Régulateur de vitesse Gain P Vitesse d'adaptation inférieure	
D p1461	[0] ---	40.0 %	Drehzahlregler Kp Adaptiondrehzahl oben Skalierung / Speed controller Kp adaptation speed, upper scaling / Régulateur vitesse Gain Kp Vitesse d'adapt. supér.	
D p1462	[0] ---	22.00 ms	Drehzahlregler Nachstellzeit Adaptiondrehzahl unten / Speed controller integral time adaptation speed lower / Régulateur vitesse Temps d'intégration Vitesse d'adapt. infér.	
D p1463	[0] ---	600.0 %	Drehzahlregler Tn Adaptiondrehzahl oben Skalierung / Speed controller Tn adaptation speed, upper scaling / Régulateur vitesse Temps Tn Vitesse d'adapt. supér.	
D p1464	[0] ---	2'000 1/min	Drehzahlregler Adaptiondrehzahl unten / Speed controller adaptation speed, lower / Régulateur de vitesse Vitesse d'adaptation inférieure	
D p1465	[0] ---	30'000 1/min	Drehzahlregler Adaptiondrehzahl oben / Speed controller adaptation speed, upper / Régulateur de vitesse Vitesse d'adaptation supérieure	
D p1590	[0] ---	11'555 A/Vs	Flussregler P-Verstärkung / Flux controller P gain / Régulateur de flux gain P	
D p1715	[0] ---	0.530 V/A	Stromregler P-Verstärkung / Current controller P gain / Régulateur de courant Gain P	
D p1717	[0] ---	3.00 ms	Stromregler Nachstellzeit / Current controller integral-action time / Régulateur de courant Temps d'intégration	
p1810.11	---	nein/no/non	Modulator Konfiguration, Stromreglerdynamik höher / Modulator configuration, Current controller dynamics higher / Modulateur Configuration, Augmenter dynamique du régulateur de courant	

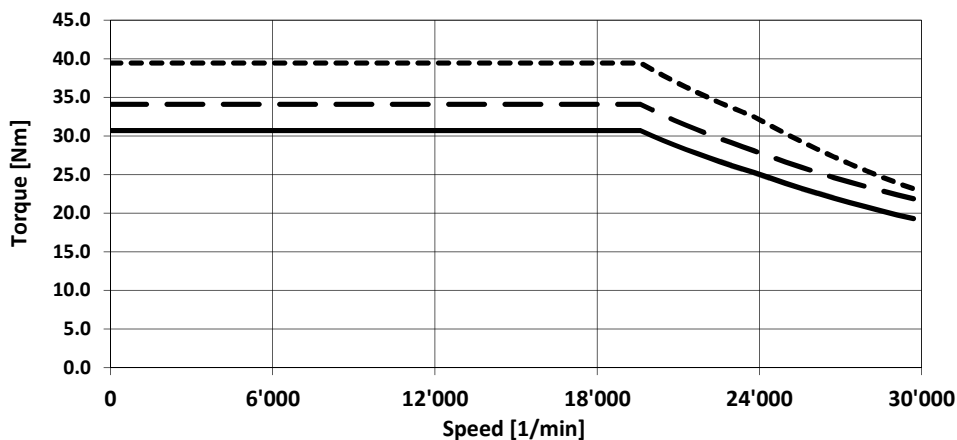
Number / Name: 106944 A SIE D 20/30 011 Index: D Issued: 12.01.2012 / HM Checked: 12.01.2012 / HM Approved: 12.01.2012 / HM Changed: 04.12.2018 / HM			Antriebsdaten Asynchronmotor drive parameters asynchronous motor paramètres moteur asynchrone			
Parameterliste Sinamics / parameter list Sinamics / liste de paramètres Sinamics						
Achseinstellungen						
35200	---	250	\$MA_GEAR_STEP_SPEEDCTRL_ACCEL (Beschleunigung im Drehzahlsteuerbetrieb)			
35220	---	0.666	\$MA_ACCEL_REDUCTION_SPEED_PPOINT (Drehzahl für reduzierte Beschleunigung)			
35230	---	0.310	\$MA_ACCEL_REDUCTION_FACTOR (Reduzierte Beschleunigung)			
35242	---	1	\$MA_ACCEL_REDUCTION_TYP (Art der Beschleunigungsreduktion)			
WICHTIG IMPORTANT Reglerparameter / Controller parameters / Paramètres du régulateur WICHTIG IMPORTANT						
Von FISCHER vorgegebene oder vom Antrieb automatisch berechnete Regelparameter für Stromregler und Drehzahlregler müssen durch eine fachkundige Person mit Hilfe der vom Antrieb zur Verfügung gestellten Messfunktionen wie Sollwertsprung und Führungsfrequenzgang überprüft und gegebenenfalls korrigiert werden.						
Parameters provided by FISCHER or automatically calculated controller parameters for current controller and speed controller must be verified and eventually adapted by an expert using the measuring functions setpoint step or frequency response of the drive.						
Les paramètres de contrôle du régulateur de courant et du régulateur de vitesse calculés automatiquement par le variateur ou prédéfinis par FISCHER doivent être vérifiés et, en cas échéant, corrigés par une personne compétente utilisant les fonctions de mesure fournies par le variateur, telles que le pas de consigne et la réponse en fréquence.						
1) WICHTIG IMPORTANT Temperatursensor / Temperature sensor / Sonde thermoélectrique du moteur WICHTIG IMPORTANT						
Die Einstellung in M p601 ist abhängig vom angeschlossenen Temperatursensor: Der angeschlossene Sensortyp ist im Geberschema der Montageanleitung definiert.						
Setting M p601 depends on the connected motor temperature sensor type: The actual connected temperature sensor type can be found in the encoder diagram of the assembly instruction.						
Le réglage dans M p601 dépend du capteur de température raccordé: Le type de capteur raccordé est défini dans le schéma du codeur des instructions d'installation.						

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Asynchronous motor (delta connection)
Power-Torque-Speed-Diagram



— Power [kW] S1 - 100% - - Power [kW] S6 - 60% / 2min
- . - Power [kW] S6 - 40% / 2min



— Torque [Nm] S1 - 100% - - Torque [Nm] S6 - 60% / 2min
- . - Torque [Nm] S6 - 40% / 2min

S1 - 100%	Synchronous speed	[1/min]	19'800	24'000	30'000	-	-	-
	Asynchronous speed	[1/min]	19'600	23'800	29'700	-	-	-
	Frequency	[Hz]	660	800	1'000	-	-	-
	Motor voltage	[V]	380	380	380	-	-	-
	Current	[A]	128.0	122.0	114.0	-	-	-
	Power	[kW]	63.0	63.0	60.0	-	-	-
	Cosφ (without choke)	[-]	0.790	0.840	0.860	-	-	-
	Torque	[Nm]	30.7	25.3	19.3	-	-	-
	-	-	-	-	-	-	-	-
S6 - 60% / 2min	Synchronous speed	[1/min]	19'800	24'000	30'000	-	-	-
	Asynchronous speed	[1/min]	19'600	23'800	29'700	-	-	-
	Motor voltage	[V]	380	380	380	-	-	-
	Current	[A]	140.0	134.0	131.0	-	-	-
	Power	[kW]	70.0	70.0	68.0	-	-	-
	Torque	[Nm]	34.1	28.1	21.9	-	-	-
	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-
S6 - 40% / 2min	Synchronous speed	[1/min]	19'800	24'000	30'000	-	-	-
	Asynchronous speed	[1/min]	19'600	23'800	29'700	-	-	-
	Motor voltage	[V]	380	380	380	-	-	-
	Current	[A]	159.0	154.0	140.0	-	-	-
	Power	[kW]	81.0	81.0	72.0	-	-	-
	Torque	[Nm]	39.5	32.5	23.2	-	-	-
	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-
	-	-	-	-	-	-	-	-

Remarks

- The above stated motor performance can only be achieved using the correct drive and filter type (choke, LC-filter)

10.7 Repair Receiving Form

The form is on the following page.

Datum / date / date / data

Firma / Company / entreprise / azienda			
Kontaktperson / contact person / personne de contact / persona di contatto			
Strasse / street / rue / strada		Nr. / No.	
Stadt / city / ville / città		PLZ / postal code / code postal / CAP	
Land / country / pays / paese			
Telefon		Fax	
E-Mail			
Spindelbezeichnung / spindle designation / désignation de broche / designazione di mandrino			
Serie Nummer / serial number / no. de série / no. di serie			
Problembeschreibung / problem description / description du problem / descrizione di problema			
Bearbeitungsprozesses / process of machining / processus d'usinage / processo di lavoro			
Priorität / priority / priorité / priorità	Express	JA / YES / OUI / SI	
Anmerkungen / annotation / annotation / notazione		NEIN / NO / NON / NO	

Versandhinweise / indication for transport / indication de transport / indicine di trasporto

Lieferanschrift / provide / fournisseur / fornire	Zoll / duty / douane / dogana
Fischer AG Präzisionsspindeln Ernst-Fischer-Weg 5 CH-3360 Herzogenbuchsee	MwSt-Nr. 102.213.056 Zollkonto „ZAZ“ 8821-1 Warennummer Zolltarif T-No. 8466.1000 EORI-Nr. DE1766228



HINWEIS / NOTE / NOTE / NOTA

Verwenden Sie bitte die Original-Transportkiste!
Please use the original carrier box!
Employé si vous plait l'emballage d'origine de transport!
Utile per favore la cassa originale di trasporto!



HINWEIS / NOTE / NOTE / NOTA

Pro-Forma-Rechnung mit Ursprungserklärung der Lieferung beilegen!
Enclose the Pro forma invoice with the certified declaration of origin in the shipment!
Ajouté la facture pro forma avec la déclaration d'origine à la livraison!
Accludere la fattura pro forma con la dichiarazione d'origine con la fornitura!

10.8 Checklist for Installation & First-Time Operation

The checklist is merely an additional aid for the installation and first-time operation.

Please refer to the content of the Operating Instructions preceding the Checklist.

Spindle Installation

- The spindle sleeve, was it cleaned and greased? ☐
- The spindle support on the machine, was it cleaned and greased? ☐
- The spindle fixing bolts, were they tightened with the torque as required? ☐
- All spindle conduits and connections, were they installed according to the inscriptions and diagrams in Chapter 10: Annex. ☐
- All tubes, do they fit correctly in the corresponding Legris fittings? ☐
- The tubes, are they free of bends or bucklings in any position of the spindle? ☐
- The quality of the tubes, is it sufficient for all the required pressures? ☐
- The reference or the zero point of the spindle, was it checked, adjusted and reprogrammed? ☐

Spindle Motor Cooling System

- Do you use ready-for-use coolant COOL-X? ☐
- The cooling tubes, are they correctly connected? (entry = WATER IN/Wasser ein, exit = WATER OUT/Wasser aus) ☐
- The maximum pressure of the cooling water, is it below 6 bar? ☐
- The flow control device, does it work and is it adjusted correctly? ☐
- The coolant entering the spindle, is it cooled down to the advance temperature of 20 - 25° C? ☐

Pneumatic System

- The compressed air, does it correspond the required quality as per Technical Specifications in Chapter 9? ☐
- The air, is it filtered in two steps: coarse filter 5 μm , fine filter 0.5 μm ? ☐
- The air treatment unit, has it an automatic release for the waste water? ☐
- The dynamic pressure P2 of the protecting air, was it adjusted correctly acc. to Technical Specifications in Chapter 9? (The dynamic pressure is measured about 15 cm in front of the spindle fitting) ☐
- The dynamic pressure P2 of the taper blow-out air, was it adjusted correctly acc. to Technical Specifications in Chapter 9? The dynamic pressure is measured about 15 cm in front of the spindle fitting. ☐

Oil-Air Lubricating System

- The Installation of the periphery units of the oil-air lubrication system, was it carried out correctly? (Oil-Air Lubricating Unit, Dosage Unit, Oil-, Air- and Oil-Air-Tubes) ☐
- The oil-air lubricating unit, was it correctly adjusted and/or programmed acc. to Technical Specifications in Chapter 9? (prelubricating time, interval time, pump delay time) ☐
- The quality of the recommended and used lubricating oil, does it have the purity of 15/13/10 acc. to Technical Specifications in Chapter 9? ☐
- The new or replaced dosage unit, has it been flushed with oil-air mixture for 5 - 6 hours? ☐
- The oil supply tube to the dosage unit, has it been bled properly? (dosage unit, oiltubes, etc.) Absolutely no bubbles must appear! ☐
- Dynamic Pressure P2 on the spindle, is it set correctly? ☐
- The oil tubes, do they fit properly in the Legris fitting? ☐
- The oil tubes from the dosage unit to the spindle, are they transparent? ☐
- The oil tube between the oil-air lubrication unit and the dosage unit, is it transparent and can it stand the maximum working pressure as given in Tube Specifications in Chapter 9? ☐
- The dosage unit, is it equipped with a dosage screw 10mm³ / pulse? (It is marked with a "1") ☐

- The prelubrication time, is set to or programmed 3 minutes per 1m of tube length? ☐
- The spindle, can it be started only after the prelubricating time is over? ☐
- Dynamic Pressures P2 on dosing unit for lubrication, are they all set acc. to Technical Specifications in Chapter 9? ☐
- The setting of the air pressure control switch, is it correctly adjusted acc. to Technical Specifications in Chapter 9? ☐
- The control switches of the oil-air system, do they function perfectly? (min. pressure and oillevel control switch) ☐
- The compressed air supply for the oil-air lubrication and air curtain systems, is it activated by turning on the machine main switch? ☐
- The lubricating oil in all oil tubes, is it flowing towards the spindle regularly? Important: Use a light for checking the oil flow in the transparent oil tubes! ☐

Oil Separator

- The oil separator, was it correctly positioned, ie max. 1 m over the spindle and with oil return tubes max. 3m long. ☐
- The return oil, does it flow back regularly through the transparent oil-return tubes? ☐

Toolholding Hydraulics

- The installation of the hydraulic tubes, were they correctly installed? (A/S = Tool clamping, B/L = Tool unclamping) ☐
- The hydraulic system, was it properly bled? ☐
- The hydraulic pressure, is it correctly set acc. to Technical Specifications in Chapter 9? (A/S = Clamping, B/L = Unclamping) ☐
- The toolchange, does it work perfectly? ☐
- The spindle shaft, does it rotate easily without any resistance after toolchange? ☐
- The signals of Proximity Switches S2 - S4, are they visible and properly processed? ☐
- The clamping system, can it be activated only when the spindle is in the toolchange position and the spindle shaft does not rotate? ☐

Frequency Converter

- The frequency converter, was it set according to the motor specifications in Chapter 9? ☐

Proximity Switches (Sensors)

- When Sensors S2 and S3 are activated, it should not be possible to start the spindle! (S2 = tool unclamped, S3 = clamped without tool) ☐

Thru-Coolant System (if any)

- Thru coolant, does it contain an anticorrosive agent? ☐
- Thru coolant, is it filtered? (20 - 100 µm) ☐
- Blow-out air pressure, is it 3-5 bar? ☐
- After switching off thru-coolant system, does the compressed remain under 3-5 bar pressure for about 4 - 5 seconds so as to completely blow out the remanent coolant. ☐
- During machining without coolant, is the supply tube completely discharged ie without pressure? ☐
- The leakage holes, do they face downwards in order to drain the excessive coolant. The drain tubes do they also run downwards? ☐

Tools & Toolholders

- Tools, are they properly (Quality G2.5) balanced according to the standard ISO 1940? ☐
- Tools & toolholders, are they clean? ☐
- Tools & toolholders, can you change them easily by hand? ☐
- Tools & toolholders, are they suitable for a particular machining operation? ☐
- Is the HSK toolholder (spindles with thru coolant) fitted with a coolant transfer tube as required in Paras 3.5.1 and 3.5.2? ☐
- Is the HSK toolholder (spindle without thru coolant) secured with a tube / plug as required in Paras 3.5.2 and 3.5.3? ☐

First-Time / Daily Operation

- Run-in period and / or warm-up, have they been gone through? ☐
- The procedure of the first-time operation / commissioning, has it been programmed? ☐
- The spindle, has it the ambient temperature of 20° C before being operated? ☐